Quantum Computing

Lecture 6: Quantum Error Correction, Quantum Cryptography, and Entanglement

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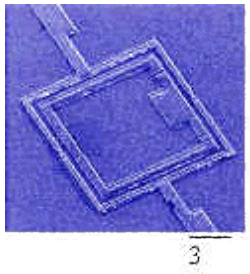
Final Exam Plan

1. After this lecture.

2. If you wrote up a solution and it is right: great!
   If you are still struggling, we will talk about the problem, together as a group.

3. These problems were designed to be HARD, (not just the last one!), so if you did make progress on trying to tackle them, then great, and if not, all I'm looking for is that you tried to conquer some of this quantum stuff (which is very different from probably everything else you've seen.)

But What Will It Look Like?



Solid State

superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

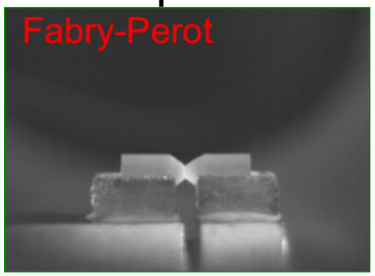
defects in diamonds

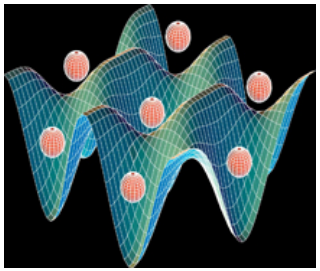
Atomic

cavity QED

neutral atoms in optical lattices

ion traps





Molecular

Liquid NMR (no longer?)

Photon Based

linear optics plus single photon devices

Pics: Mabuchi (Caltech), Orlando (MIT)

DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a scalable architecture

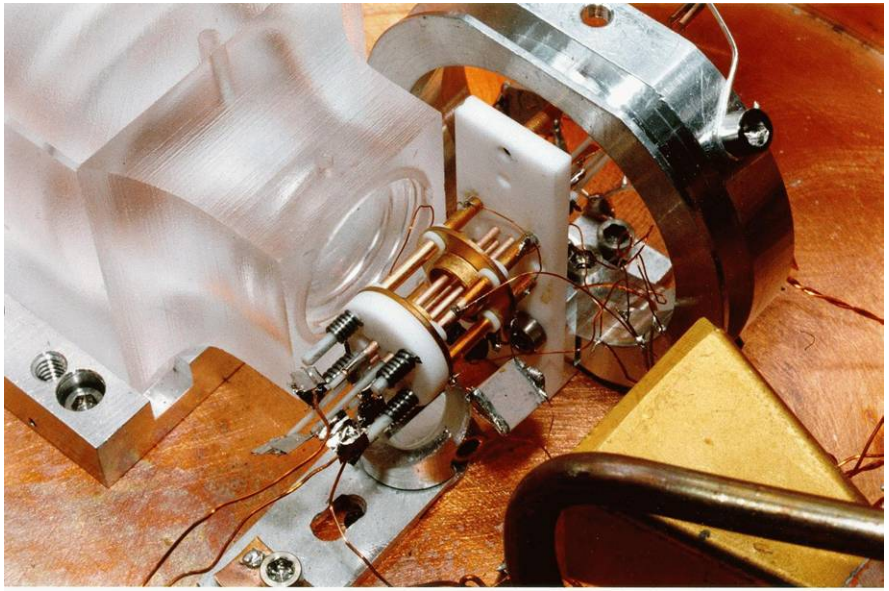
2. The ability to initialize the system to a fixed wave function.

3. Have faster control over the system than error processes in the system.

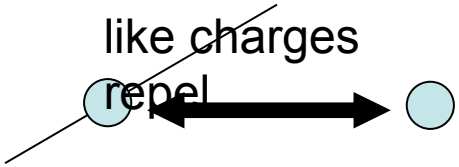
4. Have the ability to perform a universal set of quantum gates.

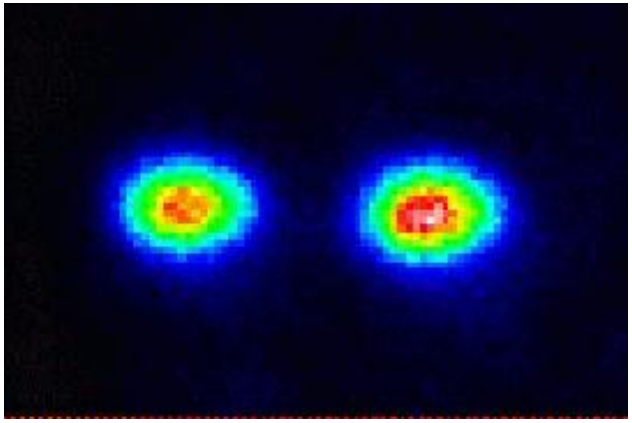
5. Have the ability to perform high quality measurements

Ion Trap

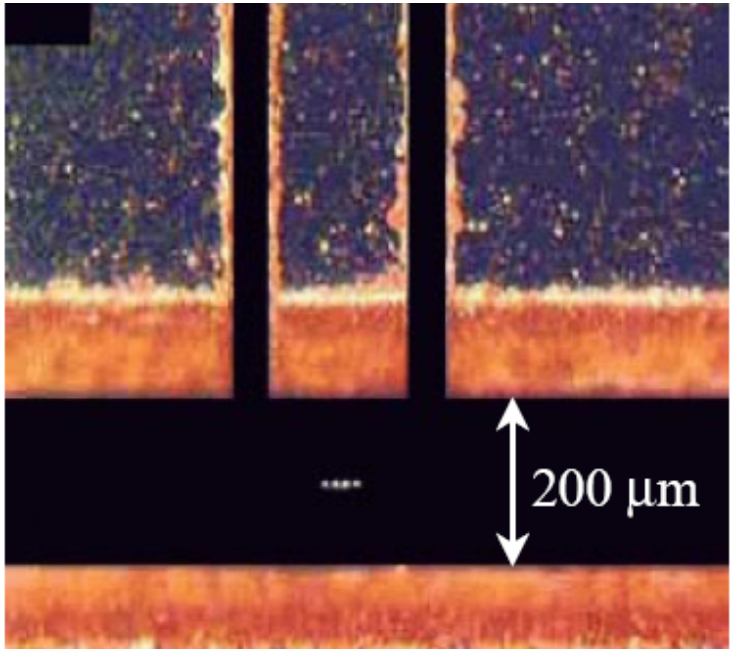


Oscillating electric fields trap ions

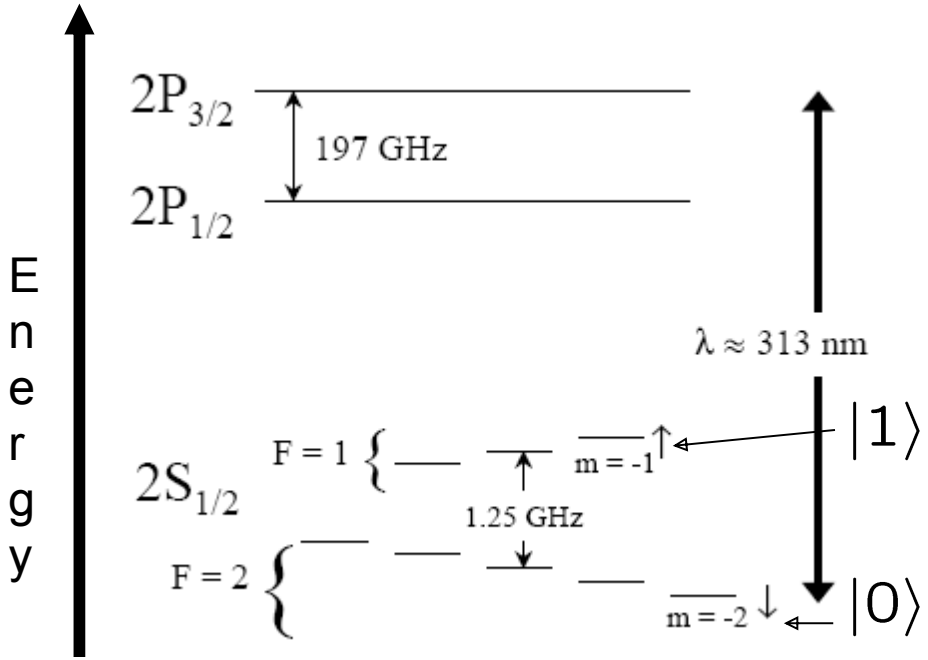


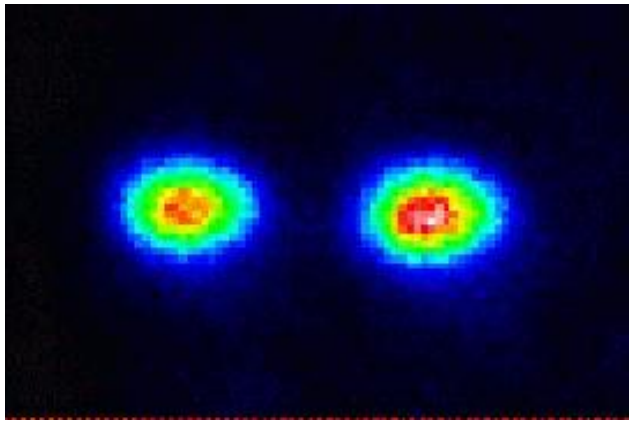


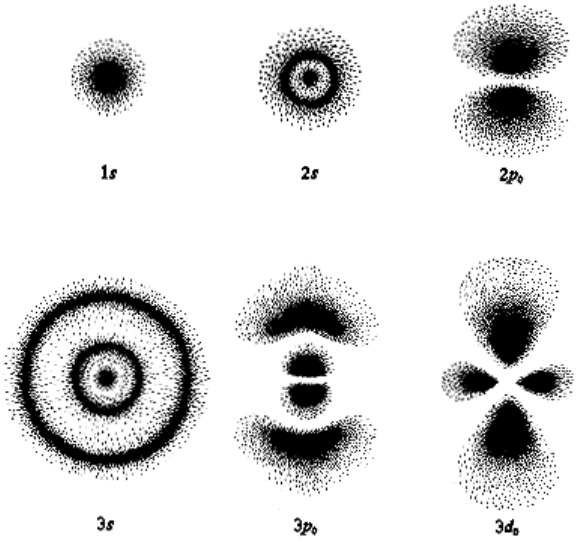
2 9Be+ Ions in an Ion Trap



Where's the Qubit?





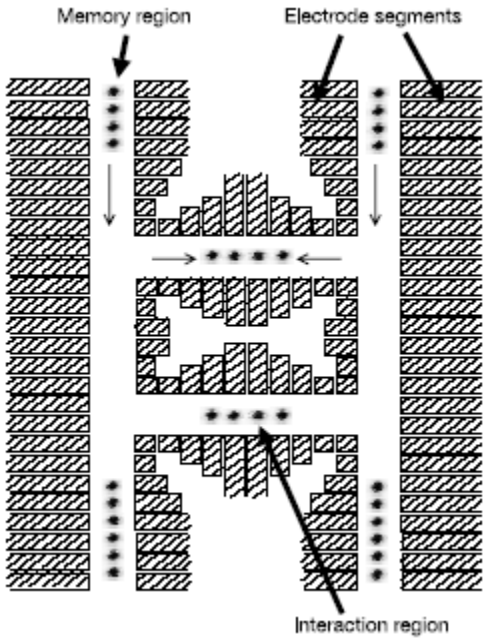


Each ion = 1 qubit

1. Well defined qubits ♠

orbitals

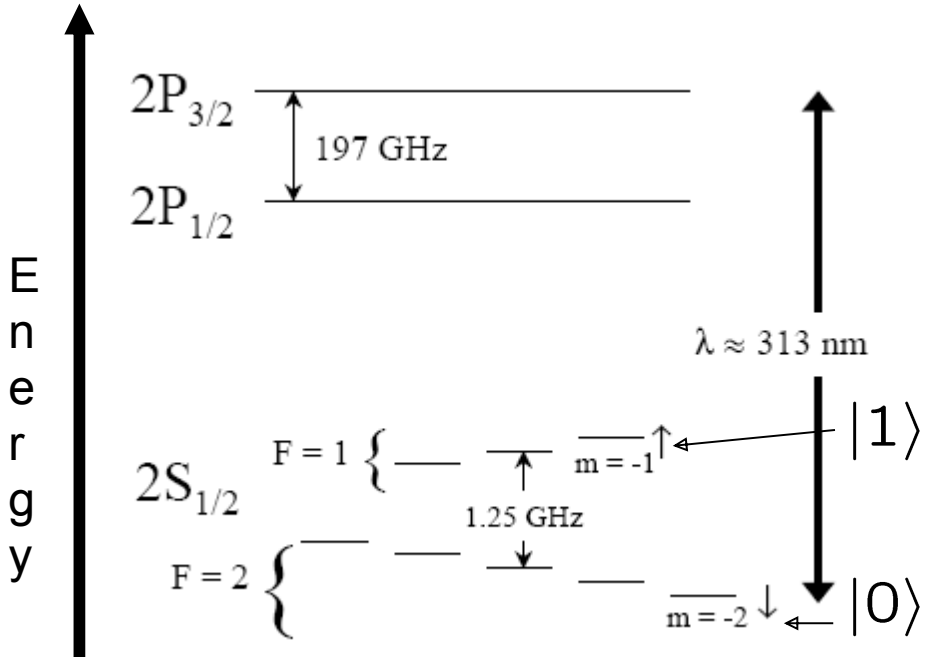
Scalable?

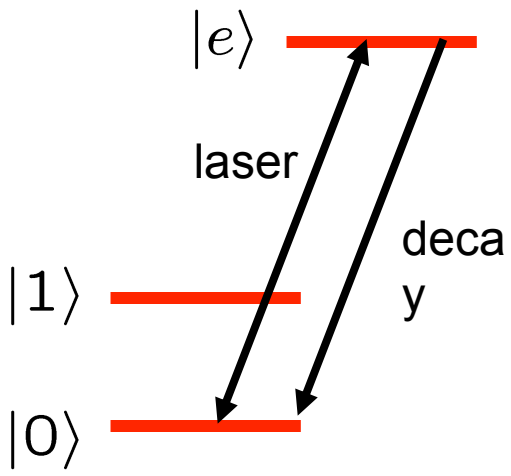


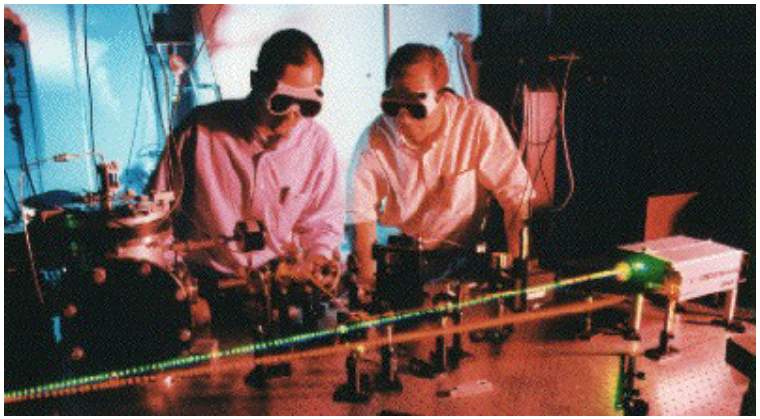
. Well defined qubits in a scalable architecture

Solid state qubits seem to have a huge advantage for scalability.

Measuremen





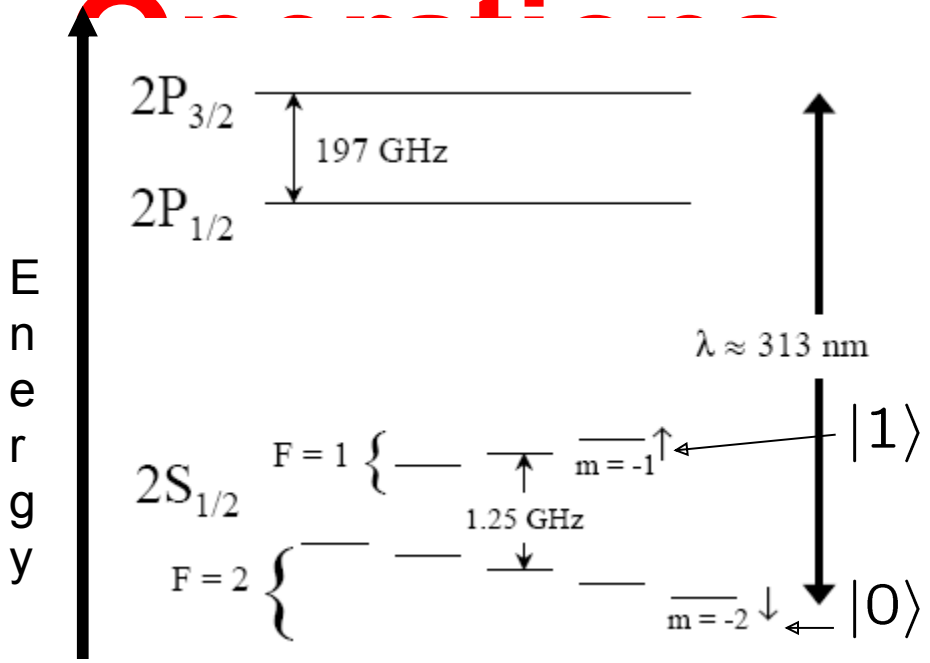


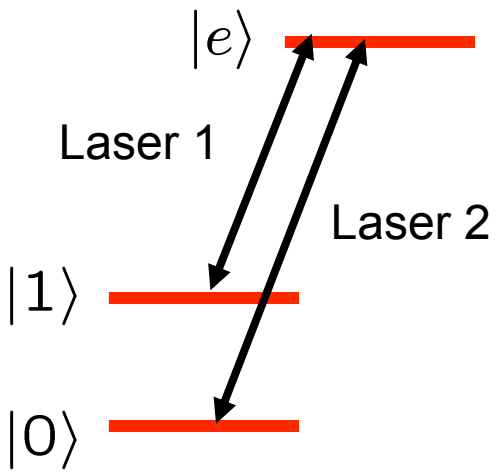
Detecting florescence implies in state 0

99.99% efficiency

5. Have the ability to perform high quality measurements ♠

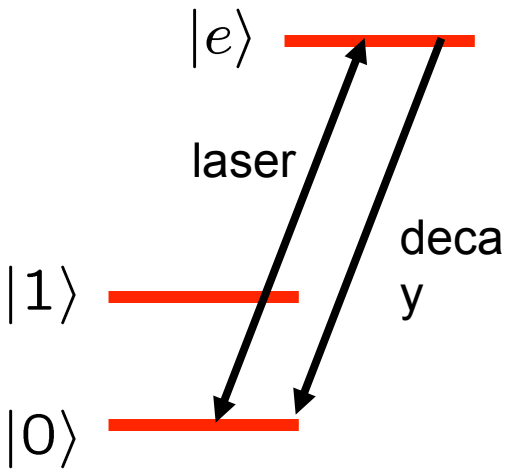
Single Qubit Operations



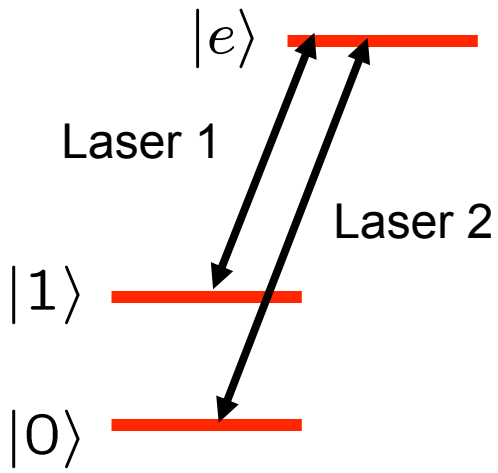


Allows any one qubit unitary operations $\begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$

Initialization



measure

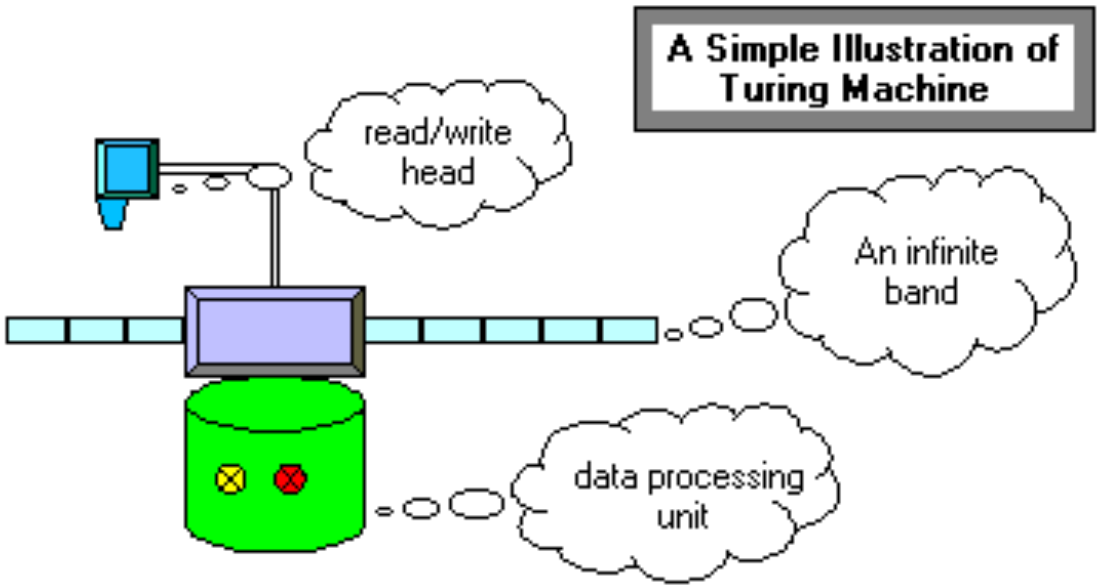


If not in zero state, flip

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. The ability to initial the system to a deterministic state. ♠

Universal C



1. Turing machine reads state of tape at current position.
2. Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right.

Certain Turing machines can perform certain tasks.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

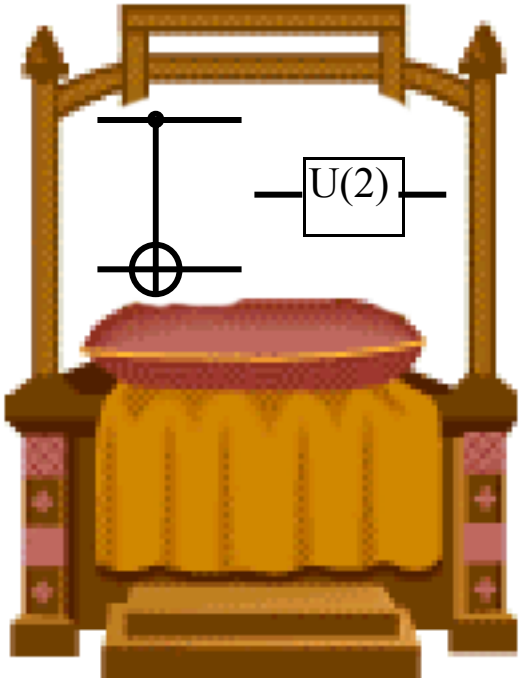
Universal Quantum Computer

Universal Quantum Computer

•a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

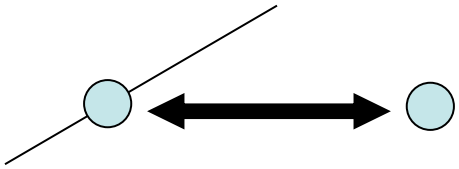
•a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



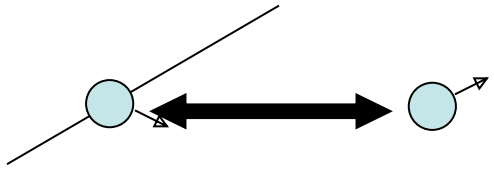
The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Coupling Two Qubits

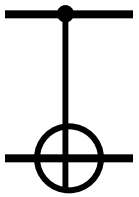


stationary



sloshing mode

These modes can be used as a bus between the qubits.



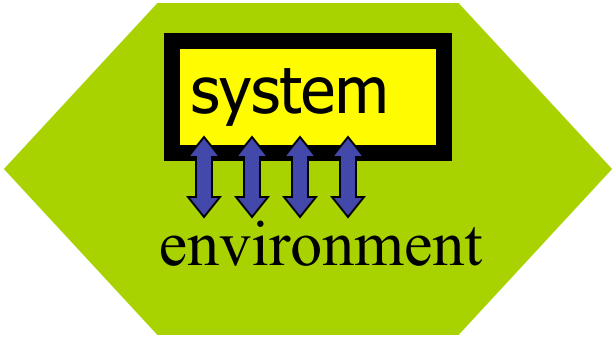
$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4. Have the ability to perform a universal set of quantum gates ♠

What is the Problem?

3. Have faster control over the system than error processes in the system.

Real quantum systems are open quantum systems!



Quantum systems readily couple to an environment…

System decoheres:

$|0\rangle$
$|1\rangle$
qubits

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

Quantum

0
1
bits

50% **0** 50% **1**

Classical

The Decoherence Problem (1996)

Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$$

Measurements are faulty

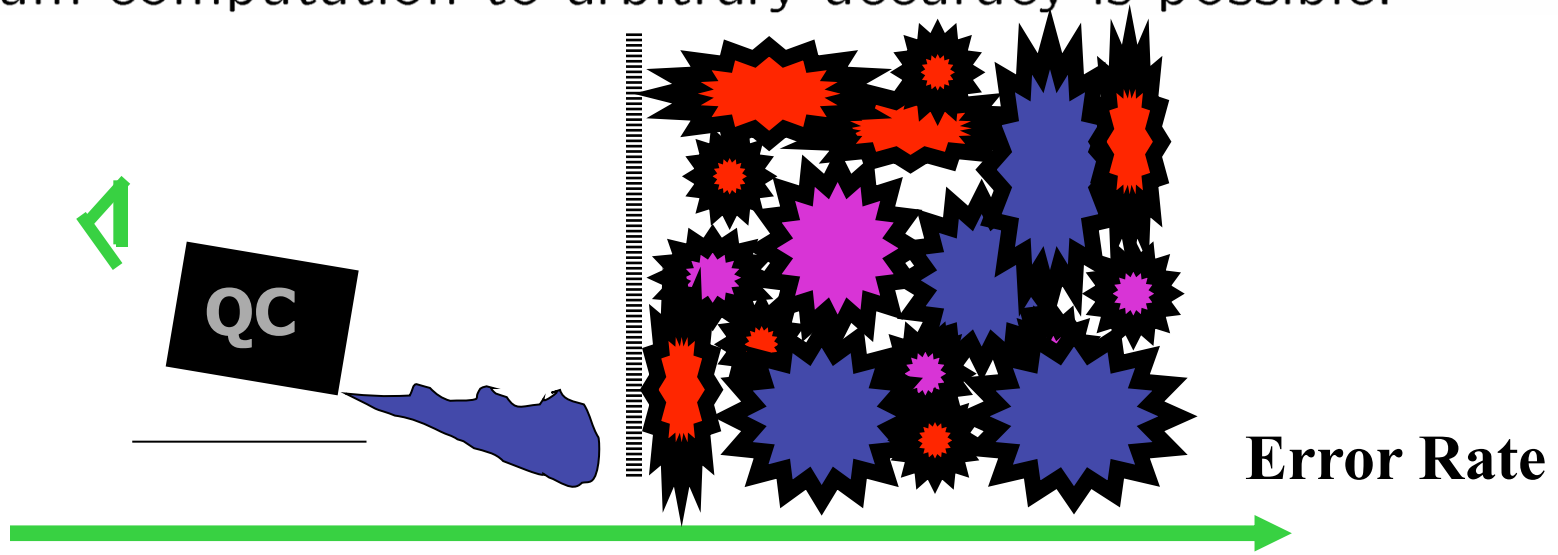
measurement result is noisy, incorrect result obtained

preparation is faulty

The Quantum Solution (1995-96)

Threshold

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.



Ion Trap Parameters

Decoherence rate for qubits: 1 minutes
Gate speed: 10 microseconds
Decoherence rate for bus: 100 microseconds to 100 milliseconds
Measurement errors: 0.01%

3. Have faster control over the system than error processes in the system. ♠

State of the Art

letters to nature

Experimental entanglement of four particles

C. A. Sackett*, D. Kielpinski*, B. E. King*†, C. Langer*, V. Meyer*, C. J. Myatt*‡, M. Rowe*, Q. A. Turchette*‡, W. M. Itano*, D. J. Wineland* & C. Monroe*

* *Time and Frequency Division, National Institute of Standards and Technology, Boulder, Colorado 80303, USA*
† *Atomic Physics Division, NIST, Gaithersburg, Maryland 20899, USA*
‡ *Research Electro-Optics, 1855 S. 57th Ct., Boulder, Colorado 80301, USA*

NIST Boulder

A Critical Ghost



Rolf Landauer
IBM

All papers on quantum computing should carry a footnote: *"This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work."*

Analog Computers

Compute by adding, multiplying real infinite precision numbers.
0.021141451124412122231112222118656…..

This can be used to solve NP complete problems in polynomial time!

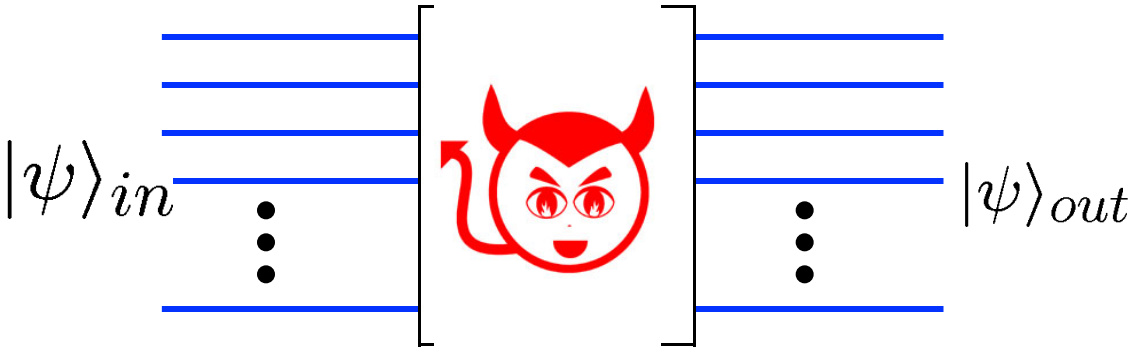
This, however is NOT a realistic model of computation.

Why? Infinite precision is requires, as far as we know, infinite resources! Noise destroys the speedup.

Is quantum computing an analog computer?

The resolution of this is the subject of quantum error correction.

Don’t Eat That Apple



plus: simple
minus: unrealistic
plus: essential ideas

Lucifer’s channel:

set of unitaries: $\mathcal{S} = \{\mathbf{U}_1 = \mathbf{I}, \mathbf{U}_2, \mathbf{U}_3, \ldots, \mathbf{U}_k\}$

probabilities: $p_1, p_2, p_3, \ldots, p_k$, $p_i \geq 0$, $\sum_i p_i = 1$

devil applies unitary $\mathbf{U}_i$ with probability p_i

no which $\mathbf{U}_i$ information available

Identity

simplest example (single qubit):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{X}, \mathbf{Z}\},\ p_i > 0$$

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

How to preserve $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?

less silly? example (two qubits):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{Z}_1\mathbf{Z}_2\},\ p_i > 0$$

$$\mathbf{Z}_1\mathbf{Z}_2 = \begin{array}{c} \begin{matrix} |00\rangle & |01\rangle & |10\rangle & |11\rangle \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

notation

$$\mathbf{P}_i = \mathbf{I} \otimes \cdots \otimes \mathbf{I} \otimes \underbrace{\mathbf{P}}_{ith\ qubit} \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Encode information into first qubit? $(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|0\rangle + \beta|1\rangle)|0\rangle = (\alpha|0\rangle - \beta|1\rangle)|0\rangle$$

Solution: encode into subspace $|00\rangle$, $|11\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|00\rangle + \beta|11\rangle) = \alpha|00\rangle + \beta|11\rangle$$

Qubit can be protected by encoding into a subspace

The Story of the Ghost



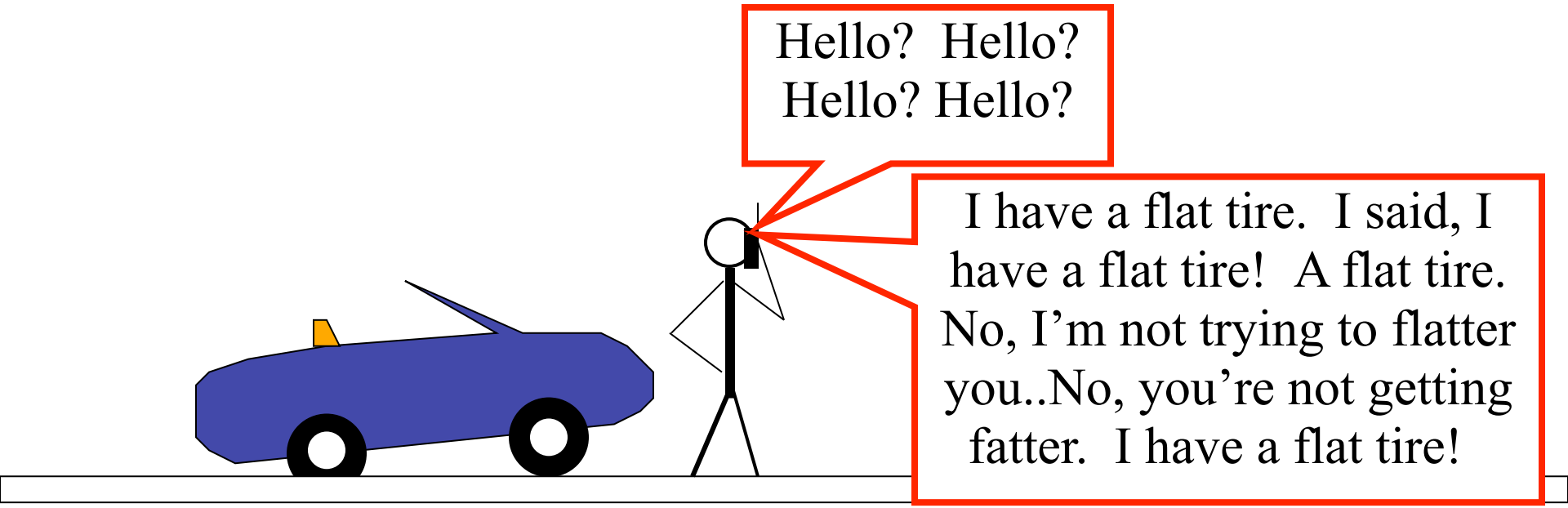


You are protecting your quantum information against a crazy noise model! Z1Z2? If this is all nature can throw at you, then pigs can fly.

Rolf Landauer
IBM

Can we use encoding to protect quantum information from more realistic noise models?

Noisy Cell Phone



Communication over a noisy CHANNEL can be overcome via

ENCODING

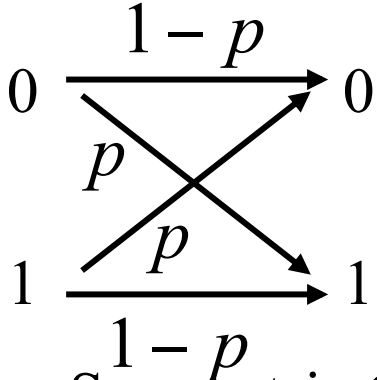
"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

Simple Repetition Code

Encode: $0 \rightarrow 000\ldots000$

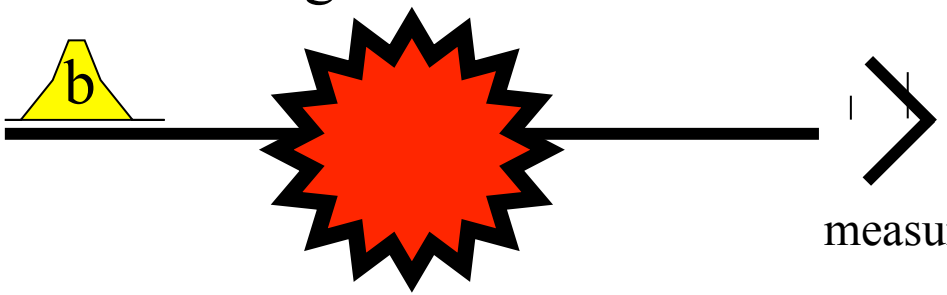
$1 \rightarrow 111\ldots111$

n copies



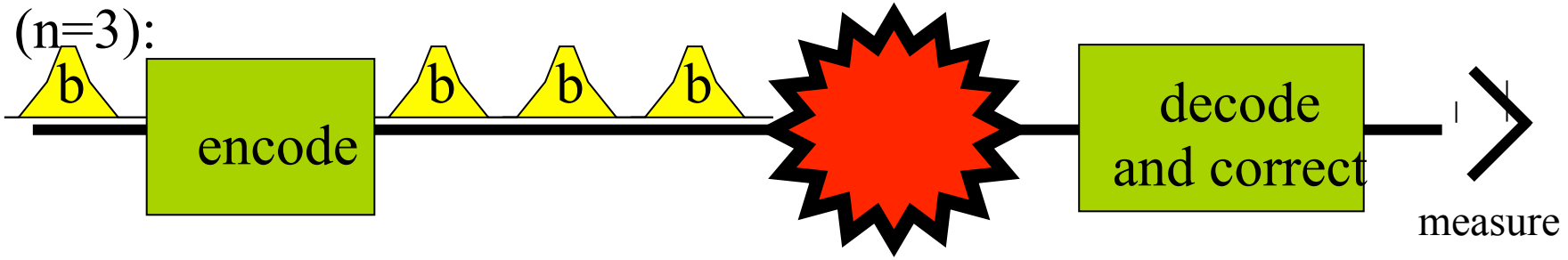
Binary Symmetric Channel

No encoding:



Probability of error = p

Encoding (n=3):

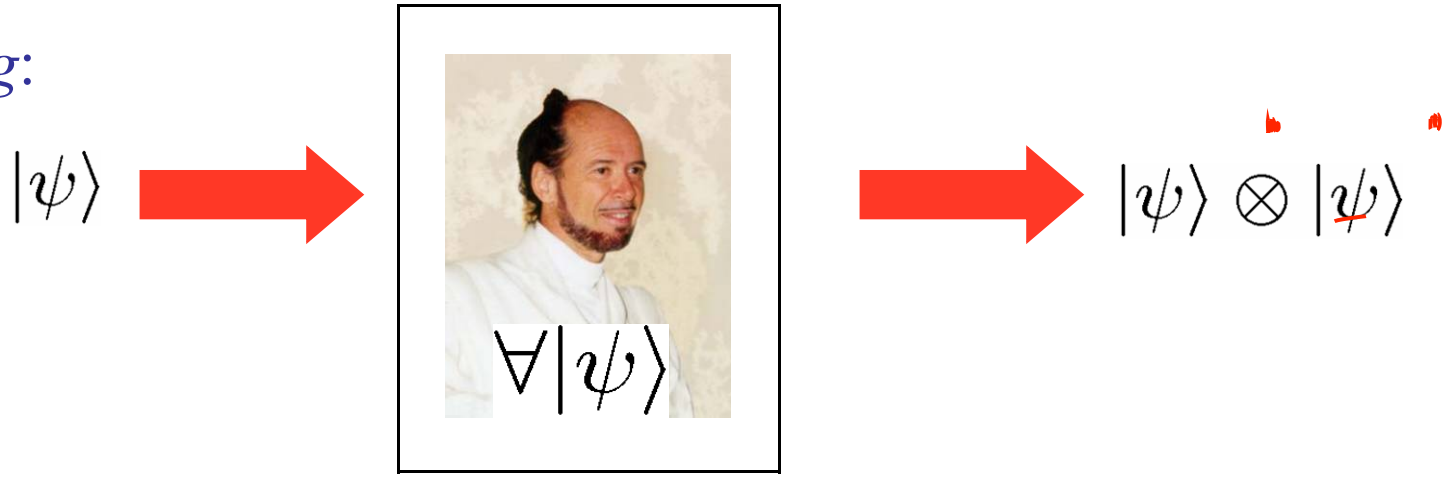


Probability of error

$$= 3p^2(1-p) + p^3 = 3p^2 - 2p^3 < p, p < \tfrac{1}{2}$$

1994 Reasons to be a Pessimist

No cloning:



Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

No quantum repetition code:

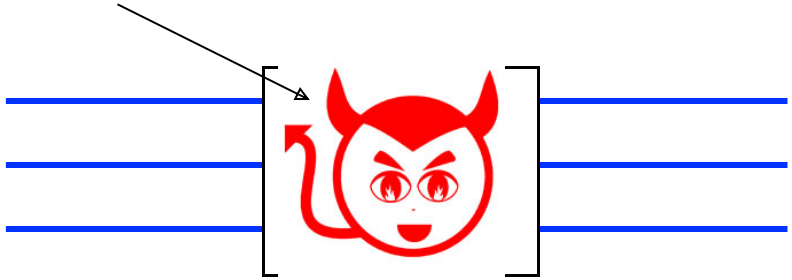
$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Measurement destroys coherence:

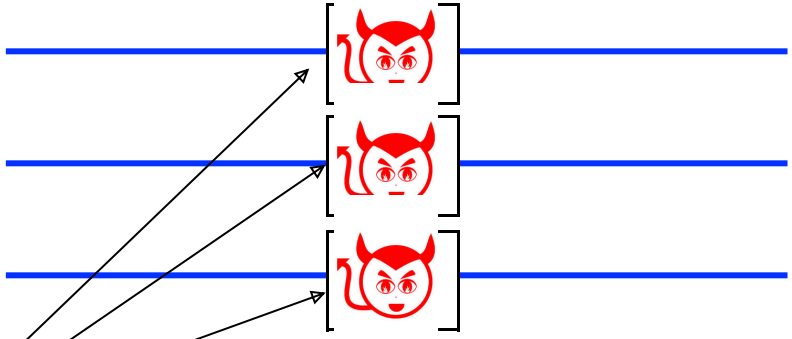
How can one decode without destroying the information?

Unrealistic Realistic Channel

$$\mathcal{S} = \{\mathrm{I}, \mathrm{X}_1, \mathrm{X}_2, \mathrm{X}_3\},\ p_1 = 1 - 3\epsilon, p_2 = p_3 = p_4 = \epsilon$$



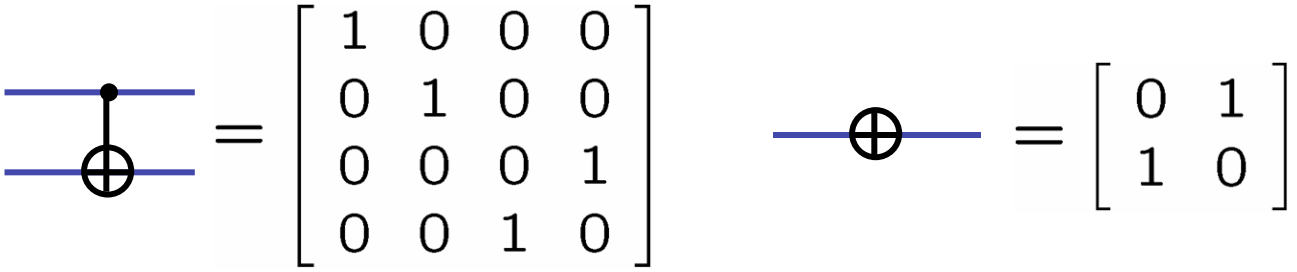
≀



$$\mathcal{S} = \{\mathrm{I}, \mathrm{X}\},\ p_1 = 1 - \epsilon, p_2 = \epsilon$$

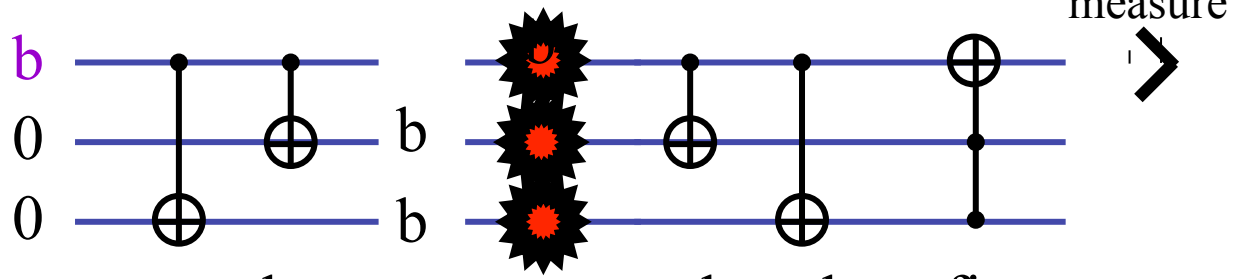
Baby Steps





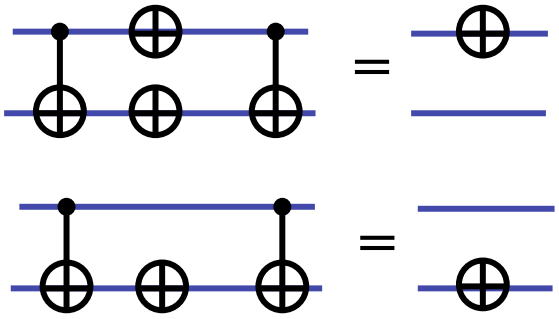
WWCCD? (What Would Classical Coders Do?)

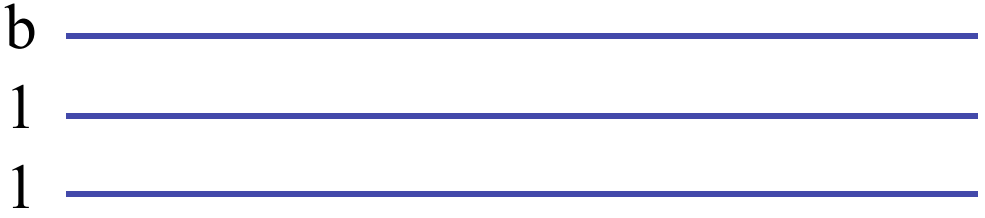
0 □000, 1 □ 111



100 □ 111□101 □ 110 □ 110

identities



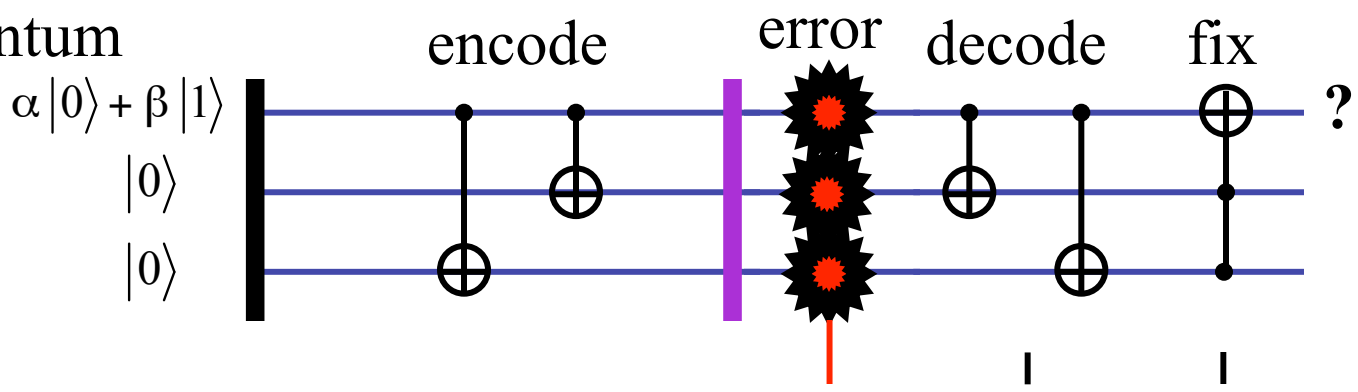


error #@%

Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

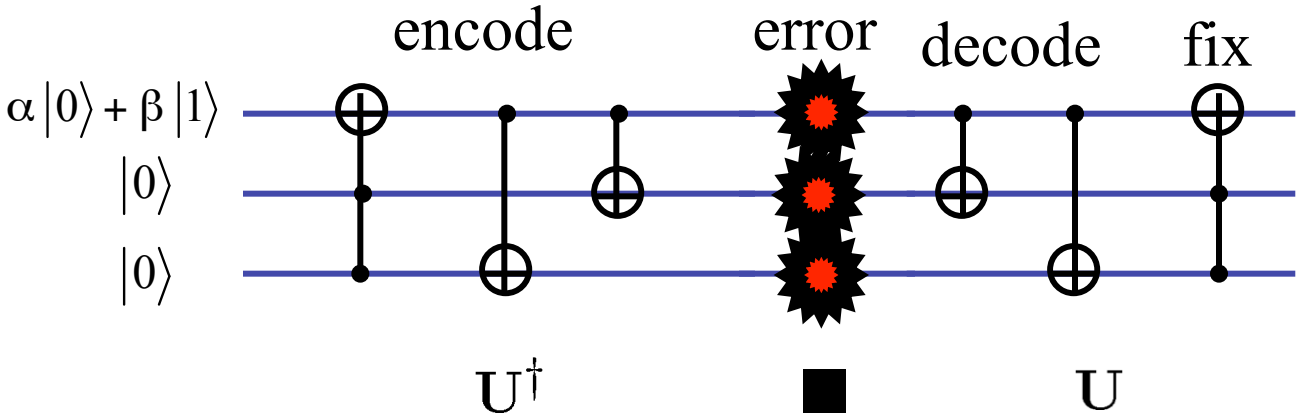
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

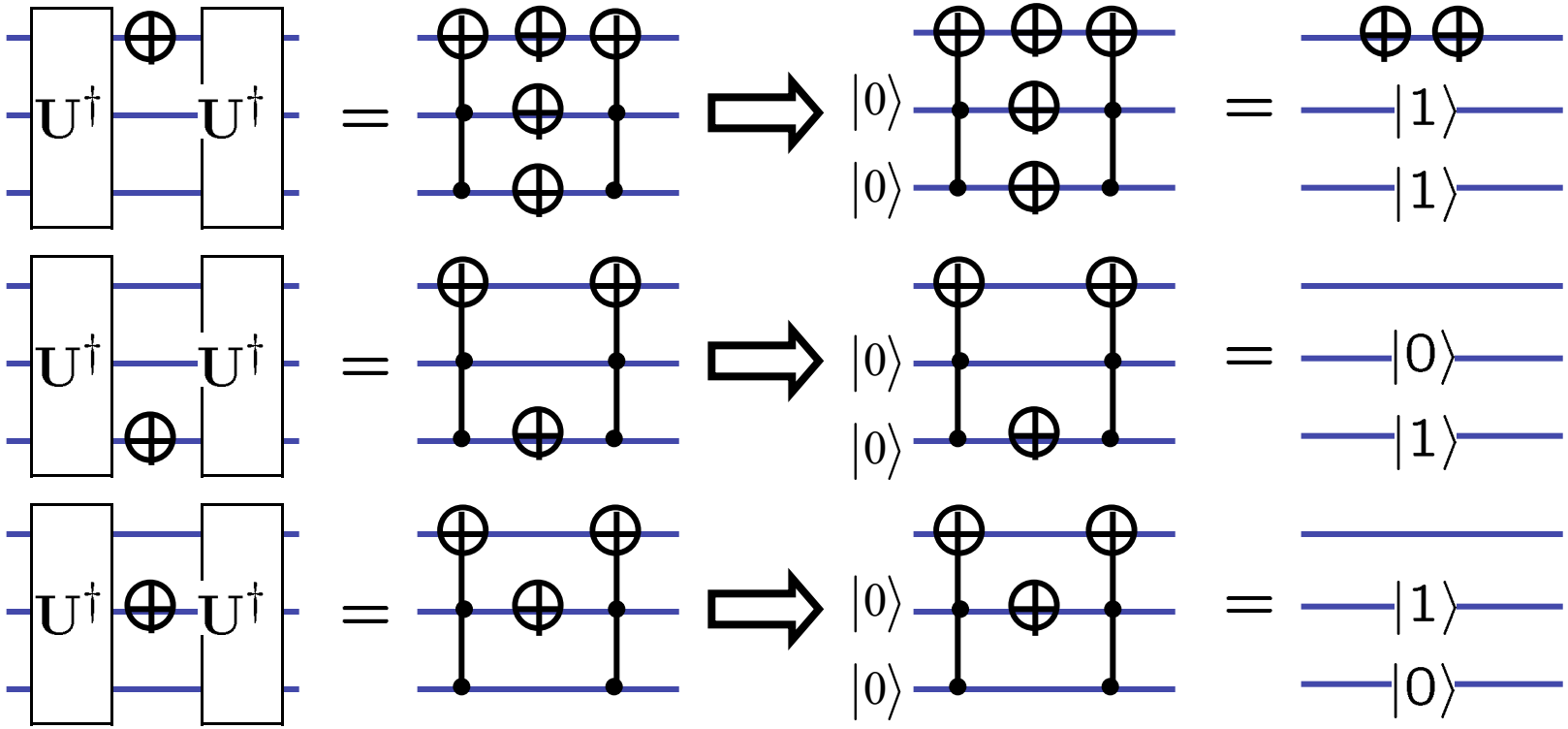
3. syndrome

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

Identity



But this is just a basis change: $UEU^\dagger$, E is the error



OK Wise Guy

What about "phase" errors?

phase error: $Z(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

$$\mathcal{S} = \{I, Z_1, Z_2, Z_3\}$$

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

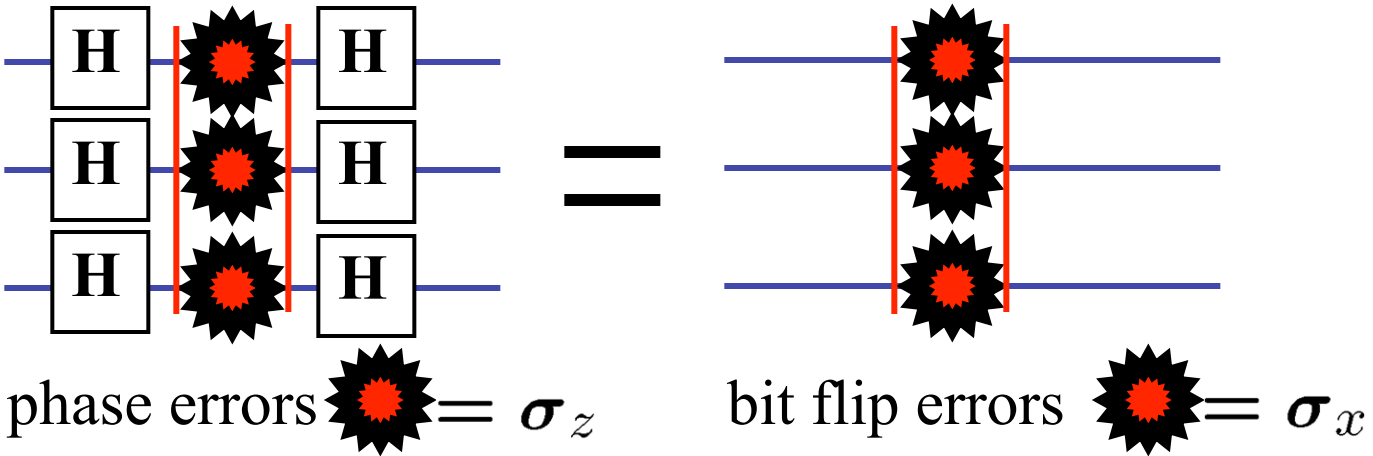
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$Z|+\rangle = |-\rangle$$

$$Z|-\rangle = |+\rangle$$

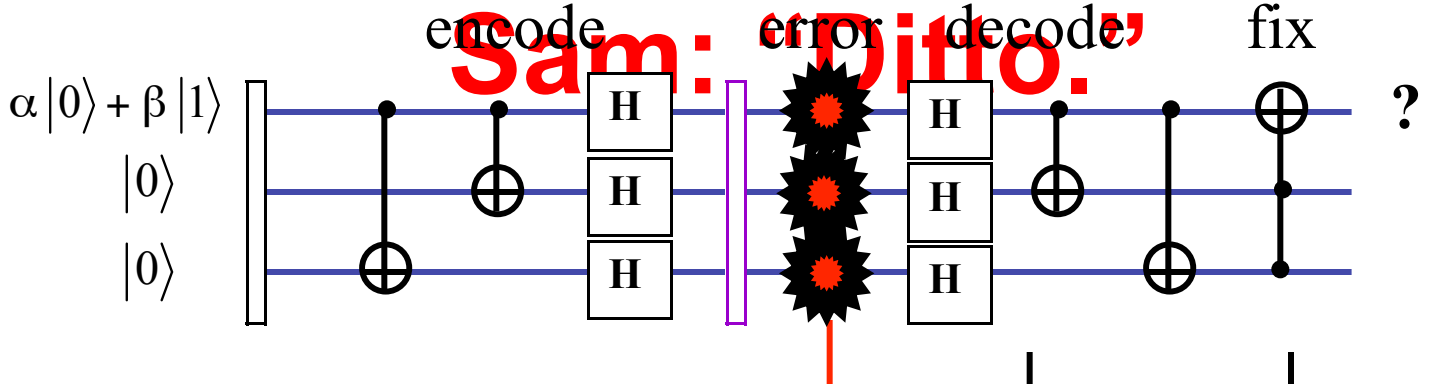
looks like bit flip error in this new basis!

$$HZH = X$$



Molly: "I love you, I really love you"

Sam: "Ditto."



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

<u>encoded into subspace</u>:

$|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

(no-cloning evaded!)

2. <u>errors take to orthogonal subspaces + maintain orthogonality</u>

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. <u>syndrome</u>

Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII**, **IZI**, **IIZ** act as ■ on bit flip code qubits:

$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

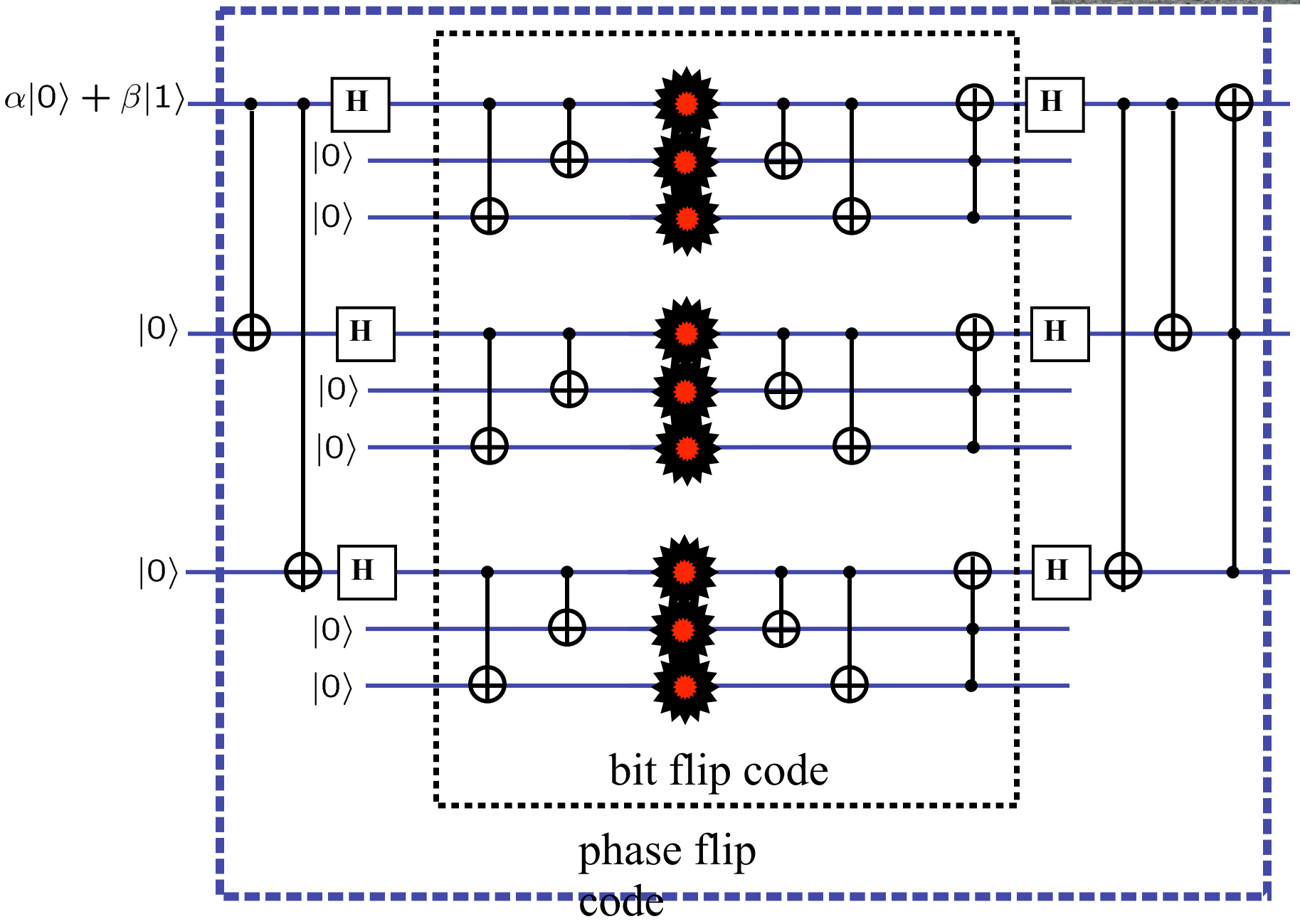
define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

Inside Shor




$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
bit flip code
phase flip
code

Linearity of Errors

We have only discussed two types of errors, bit flips and phase flips. What about “general” errors?

Theorem of digitizing quantum errors:

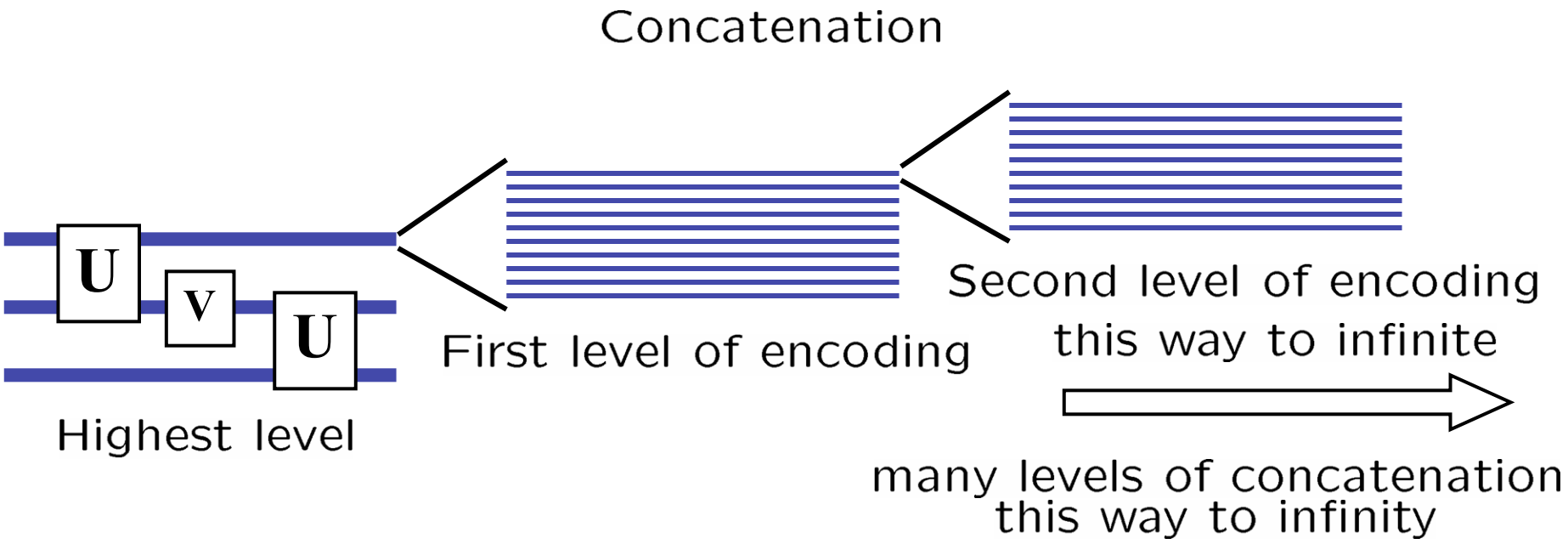
If we can correct errors in some set, $\mathcal{S}$ then we can correct any linear complex combination of such errors.

$$\text{set of unitaries: } \mathcal{S} = \{\mathbf{U}_1 = \mathbf{I}, \mathbf{U}_2, \mathbf{U}_3, \ldots, \mathbf{U}_k\}$$

While errors may form a continuous set, we only need to correct a discrete set of these errors

Perfection Through Concatenation

Idea 2: Error correct these encoded qubits to fix 'errors'



Threshold Theorem for Quantum Memory

If the single qubit error rate is below some threshold p_t, then we can store quantum states to accuracy ϵ using poly$(\log(\epsilon^{-1}))$ qubits.

Quantum Error Correction

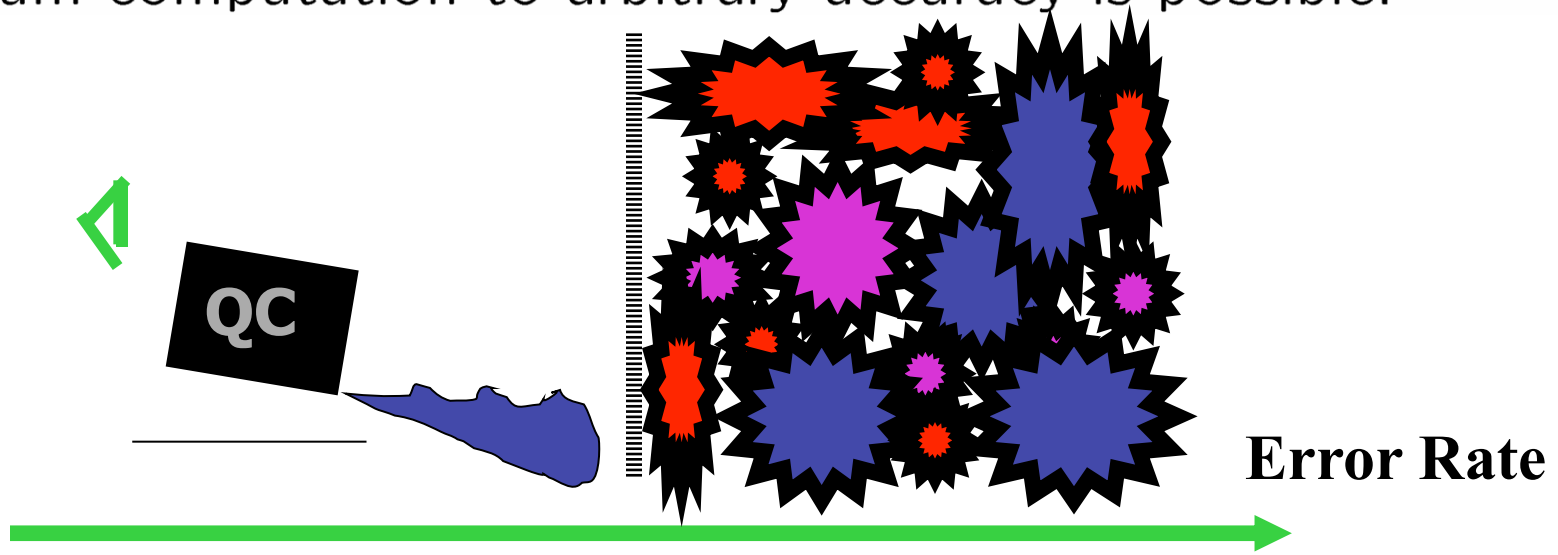
The insight that quantum computers could be defined in the presence of noise (the full theory is called fault-tolerant quantum computation) is why we have been justified in using the quantum circuit model.

Quantum error correction justifies calling a quantum computer a digital computer.

The Quantum Solution (1995-96)

Threshold

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.



Quantum Cryptography

We saw that quantum computers defeat many public key cryptosystems. Luckily quantum theory also provides an alternative, known as quantum cryptography.

Goal: a manner in which Alice and Bob can share secret key such that they can detect if an eavesdropper can be detected.

Quantum Cryptography





Alice generates 2n bits with equal probability

$$b_1, b_2, \ldots, b_n$$
$$x_1, x_2, \ldots, x_n$$

The first of these bits labels a basis choice and the second labels a wave function choice. Alice prepares n qubits:

b_i	x_i	Alice's qubit
0	0	$\|0\rangle$
0	1	$\|1\rangle$
1	0	$\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
1	1	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

Quantum Cryptography





Alice sends her n qubits to Bob.

Alice then announces via a public channel what basis she measured in: the b bitstring.

If Bob measures his qubits in the same basis, he will end up with results which exactly match Alice's bit string

$$x_1, x_2, \ldots, x_n$$

They can then reveal a few of their bits at random to check whether someone has been eavesdropping.

If not eavesdropping, the rest of their bits are a shared

Quantum Cryptography





Eve sees a procession of qubits in the computational or plus/minus basis. Eve does not know the basis.

Intuition: If Eve tries to measure this qubit, since she doesn't know what basis to measure in, sometimes she will make measurement in the wrong basis and this can be detected by Alice and Bob.

Quantum Cryptography





0	1	1	0	0
1	0	0	0	1

$|1\rangle \quad |+\rangle \quad |+\rangle \quad |0\rangle \quad |1\rangle$

b_i	x_i	Alice's qubit
0	0	$\|0\rangle$
0	1	$\|1\rangle$
1	0	$\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
1	1	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

0	0	1	1	1

Eve's basis

$|1\rangle$ 50% $|0\rangle$ 50% $|1\rangle$ $\quad |+\rangle$ $|+\rangle$ $|+\rangle$ 50% $|+\rangle$ 50% $|-\rangle$

State after Eve's measurement

Quantum Cryptography





Eve sees a procession of qubits in the computational or plus/minus basis. Eve does not know the basis.

Proof of security, with certain key generation rate, against all types of Eve's attacks.



Quantum Cryptography







- MagiQ (New York)
- id Quantique (Geneva)

Big Picture

Entanglement has long been known to be one of the
fundamentally strange things about quantum theory.

For years, people worried about entanglement.
(Einstein, Schrodinger, Bell,….)

What happening in quantum computing is that people stopped learning to worry about entanglement, and began to realize that if they just accepted it, it was a very valuable
resource!

Accept Entanglement!

But what is entanglement? Why is it “mysterious”?
Why is it important for quantum computation?

Bipartite Entanglement

$|v\rangle$



Alice's qubit

Bob's qubit



Two qubits have a wave function $|v\rangle$ which is either

Separable: we can express it as

$$|v\rangle = |a\rangle \otimes |b\rangle$$

$|a\rangle, |b\rangle$ valid wave functions

Entangled: we cannot express it as $|v\rangle = |a\rangle \otimes |b\rangle$

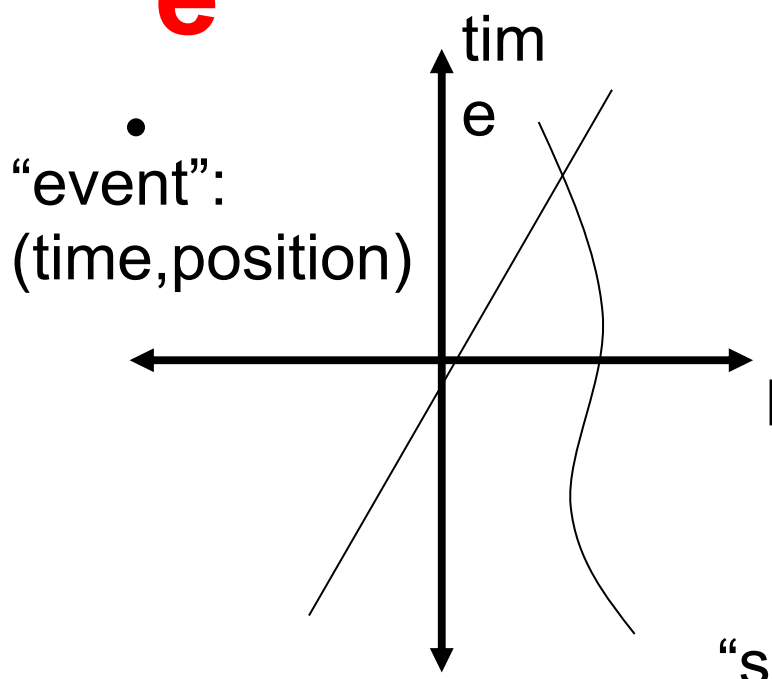
$$|v\rangle \neq |a\rangle \otimes |b\rangle$$

Special Relativity

To understand what makes entanglement so interesting in physics we need to know a little special relativity.

We don’t need to know how to calculate in special relativity, but we do need to understand the concept of “locality” which
arises in special relativity.

Spacetime



“event”:
(time,position)

“inertial frame”:
constant velocity

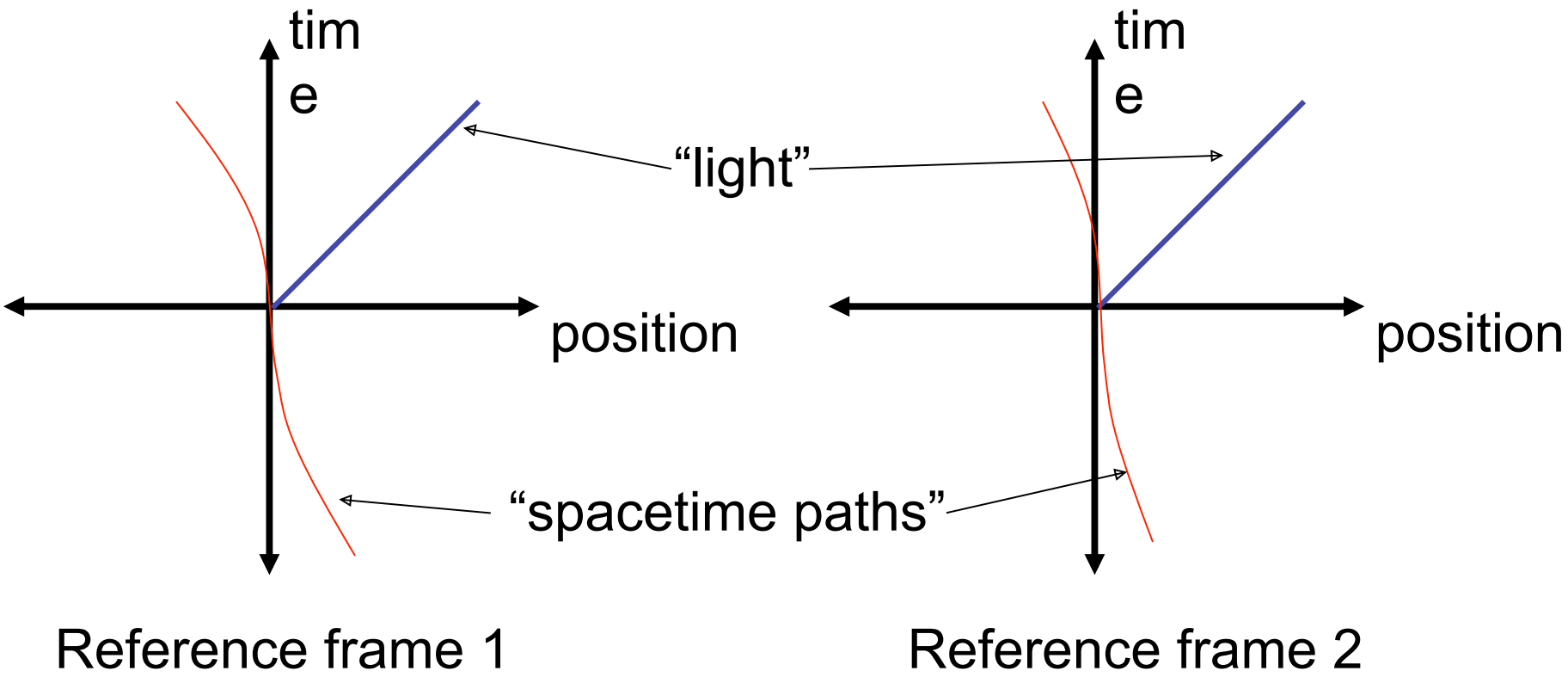
“spacetime path”:
curve in
spacetime

Special Relativity

Special relativity:

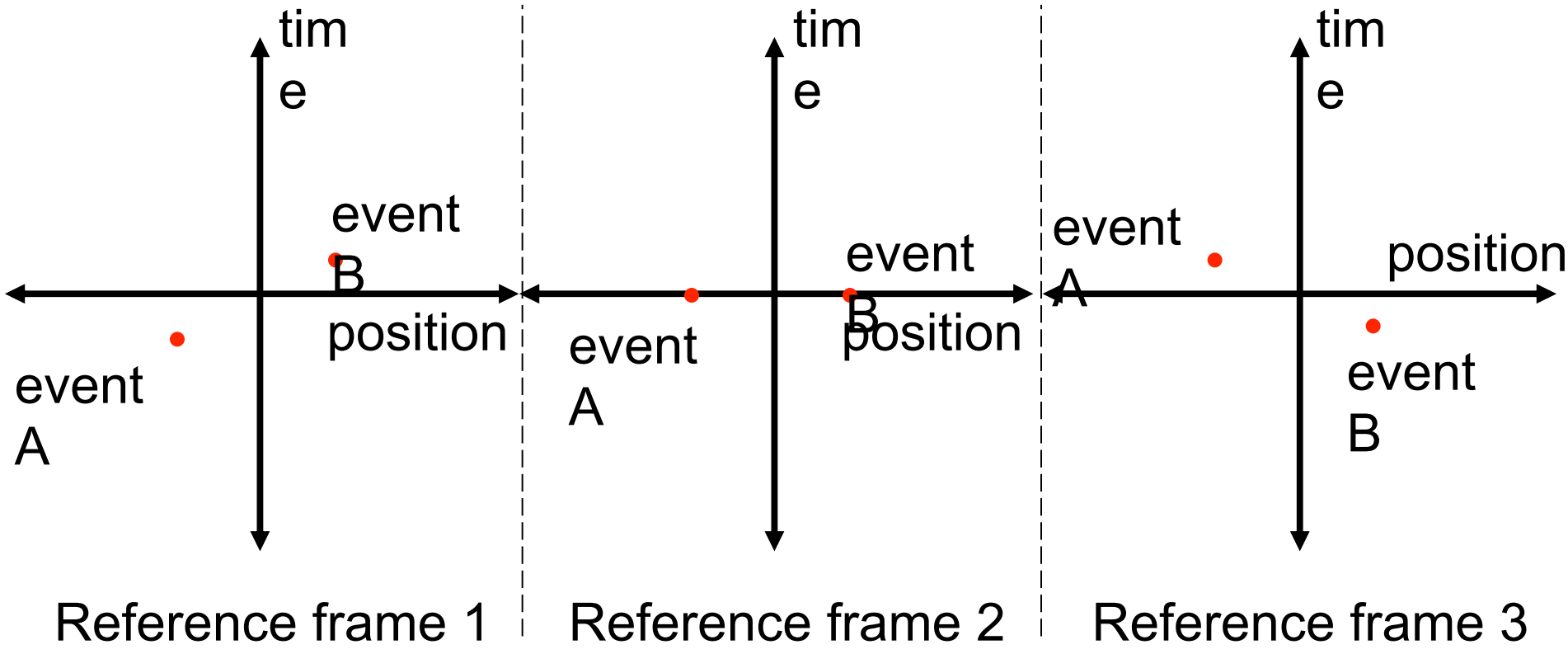
(1) physics is the same in all inertial reference frames

(2) speed of light is same in all reference frames.



Simultaneity

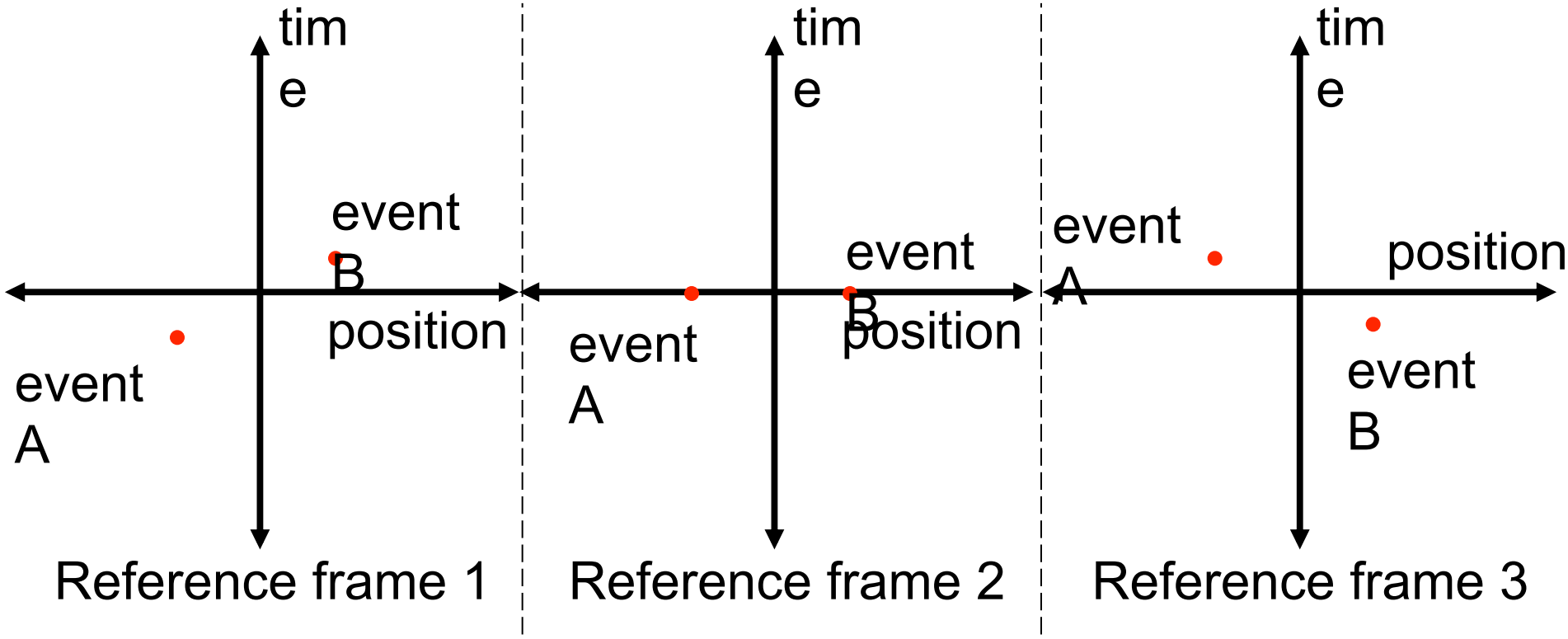
In special relativity, the idea of simultaneity is relative:



Different events are seen as occurring in a different order, depending on the reference frame.

Special Relativity

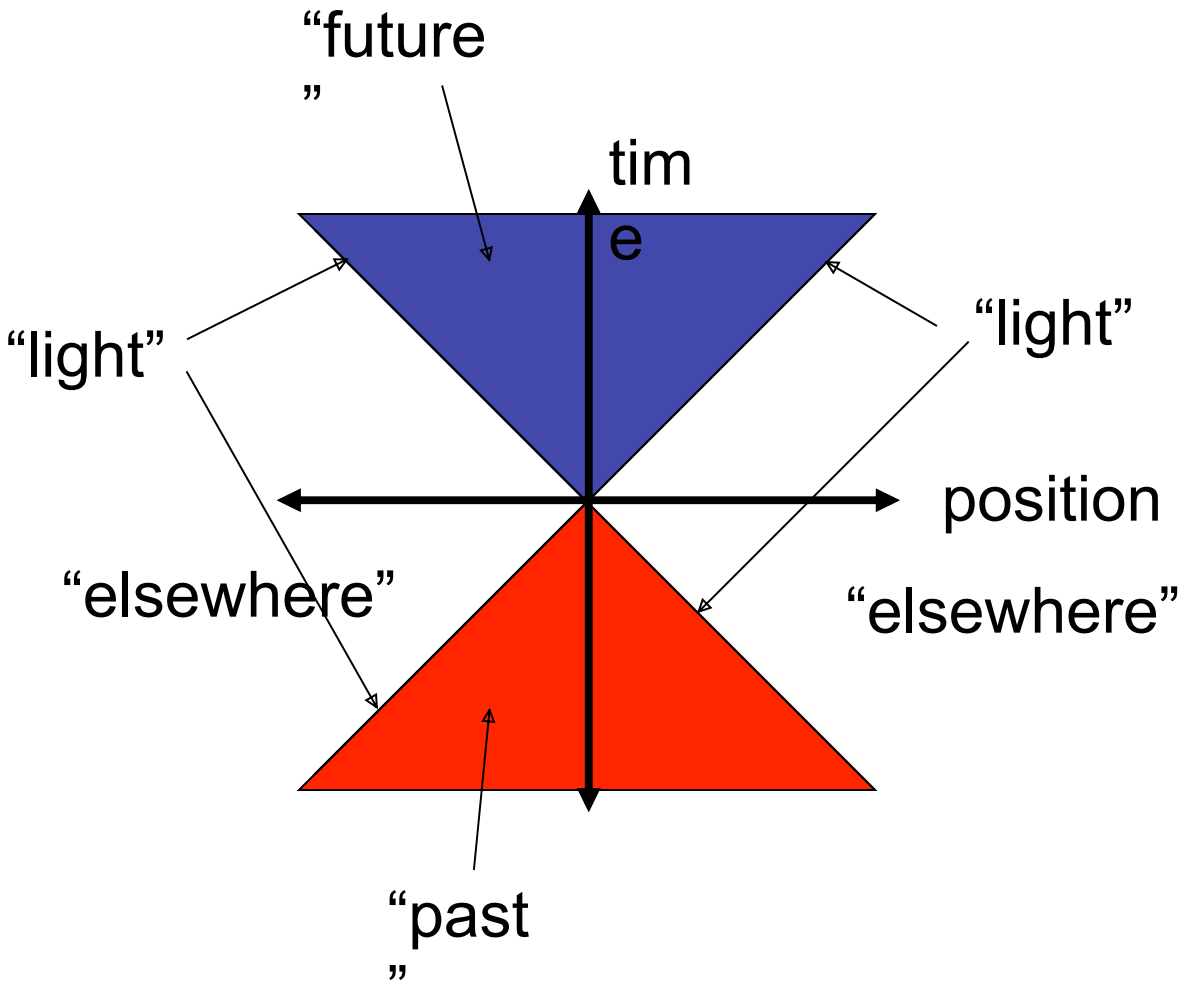
Signal: I do something at my location and time such that you can, conditional on what I do, act conditionally at your location and time.



If I try to send a signal from A to B, then in some reference frame, B acts before A sends the signal. Better not allow this!

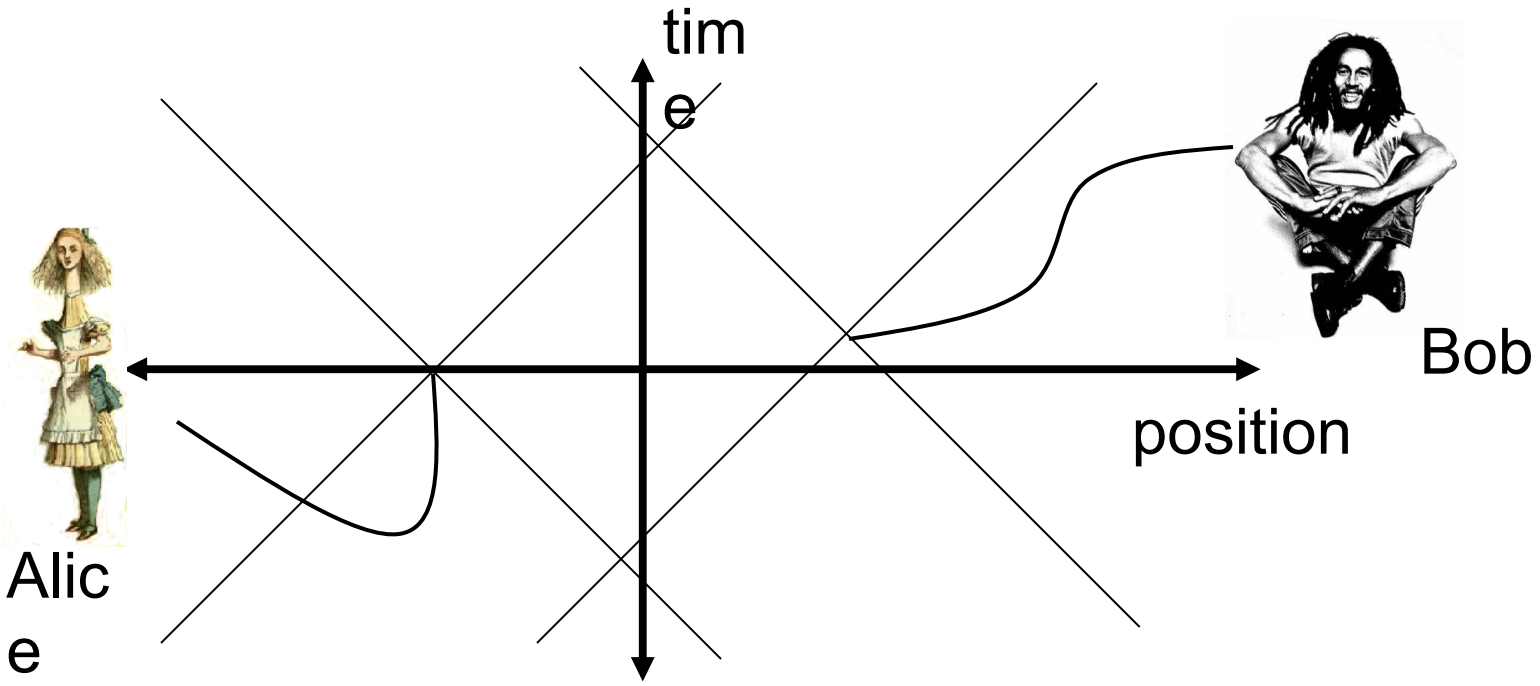
Special Relativity

For all inertial observers moving through the origin, there are three regions of spacetime which are preserved between inertial frames:



Special Relativity

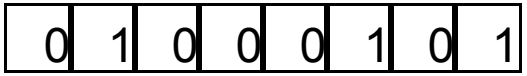
When Alice and Bob both live in each other's "elsewhere" they cannot communicate with each other:



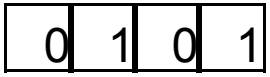
We call such setups "space-like separated"

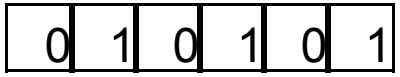
Cellular Automata

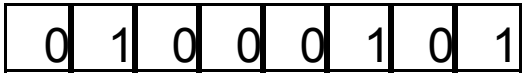
Cellular automata (CA):

State: 

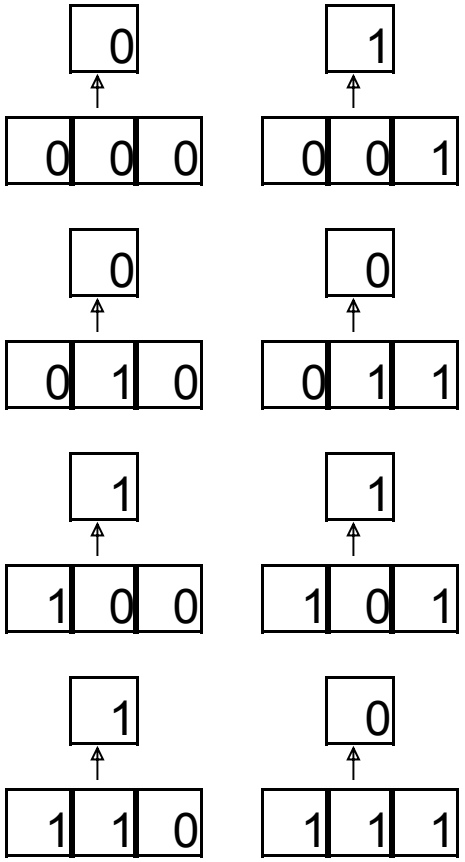
Evolution of a CA:

T= 2 

T= 1 

T= 0 

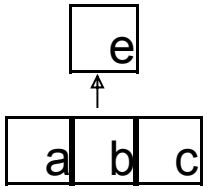
CA Rules

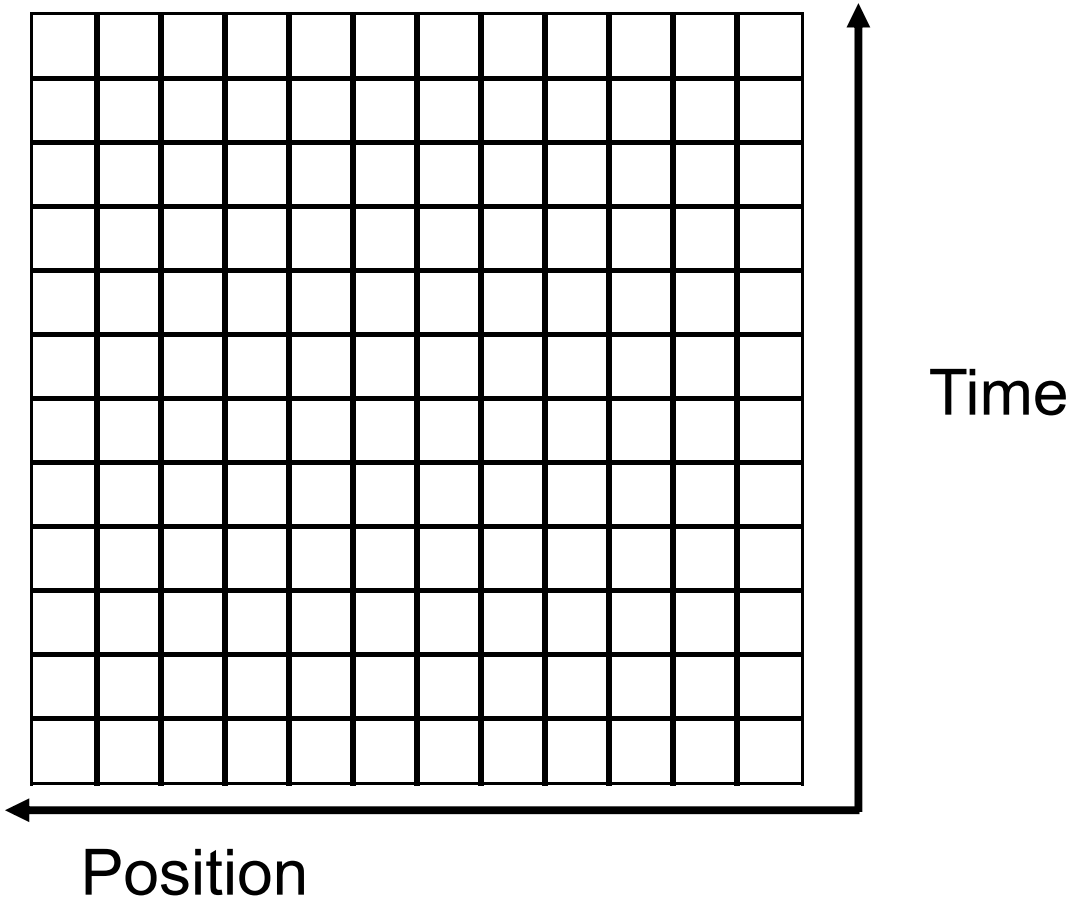


These rules are <u>local</u>

Cellular Automata

CA rules are local

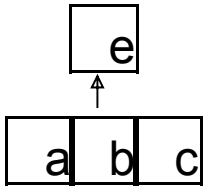




CA “Spacetime”

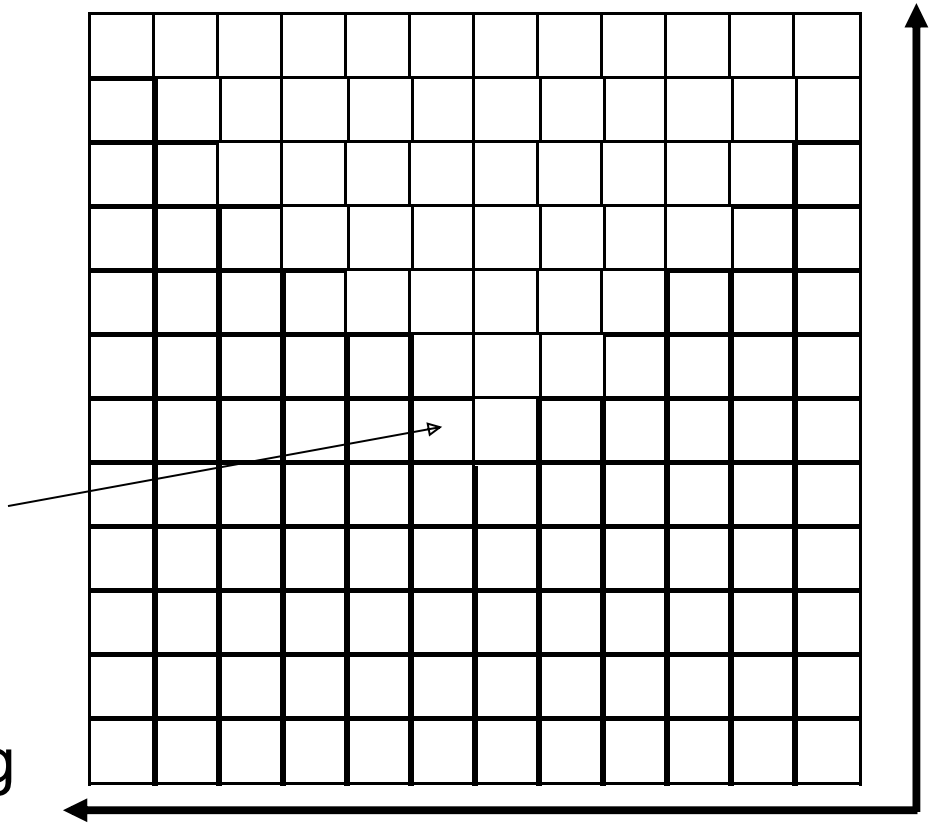
Cellular Automata

CA rules are local



Changing this value can only every change
the blue states (the future)

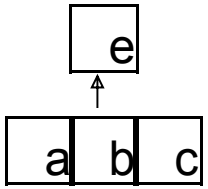
Finite speed of signaling



CA “Spacetime”

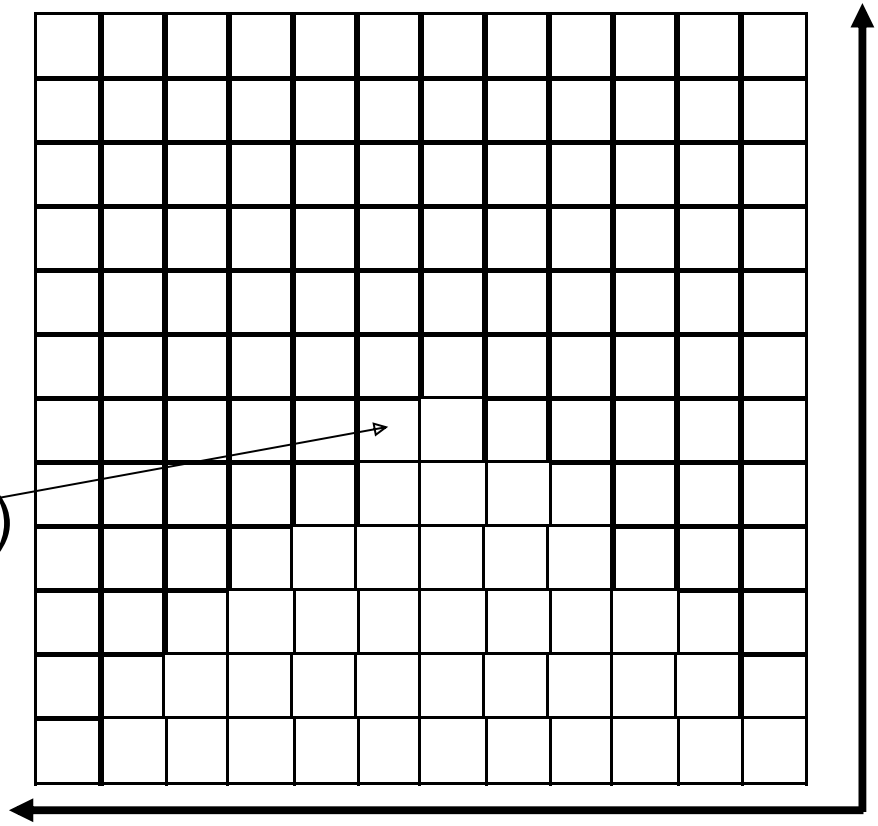
Cellular Automata

CA rules are local



This state is a function of the states in red (the past)

Finite speed of signaling



CA “Spacetime”

Bipartite Entanglement





Alice's qubit



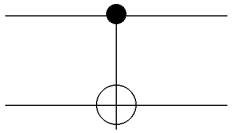
Bob's qubit

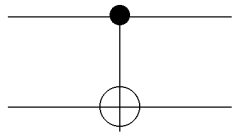


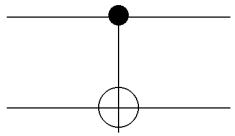
We will be interested in situations where Alice and Bob are spacelike separated. It is in these setups, where they cannot communicate, that quantum theory becomes interesting.

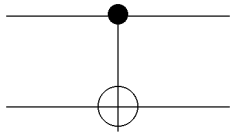
Entanglement Generation

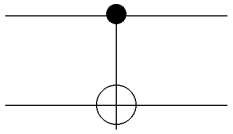
Example :



separable $|1\rangle$, $|0\rangle$  $|1\rangle$, $|1\rangle$ separable

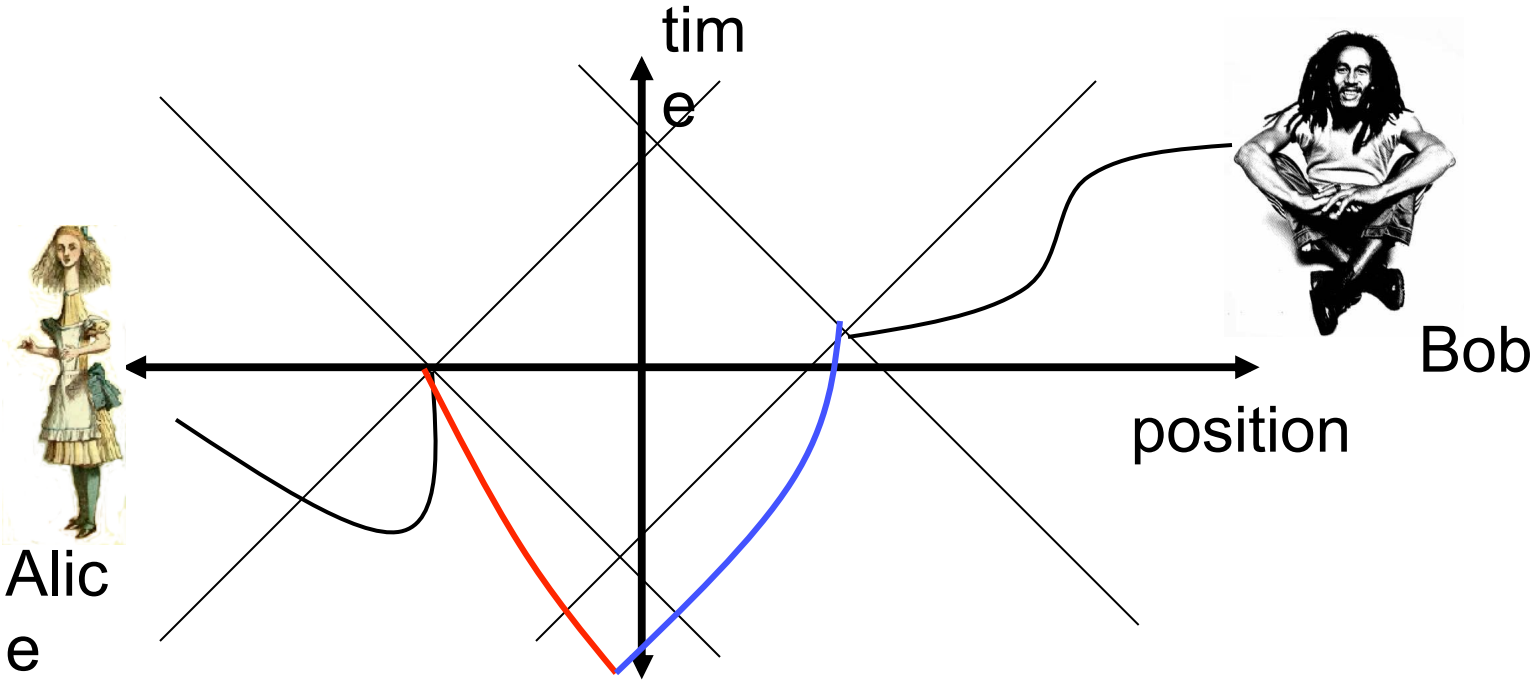
separable $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|0\rangle$  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ entangled

entangled $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$  $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|0\rangle$ separable

entangled $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$  $\frac{1}{2}(|00\rangle + |01\rangle - |10\rangle + |11\rangle)$ entangled

$\frac{1}{\sqrt{2}}(|0\rangle \otimes |+\rangle + |1\rangle \otimes |-\rangle)$ $\frac{1}{\sqrt{2}}(|0\rangle \otimes |+\rangle - |1\rangle \otimes |-\rangle)$

Entanglement Generation



Entanglement can only to be generated “locally”
i.e. the two parties got together in the past and interacted their qubits

Correlation

$|v\rangle$





$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$



Alice's qubit

Bob's qubit



1. Alice measures in the computational basis.

Facts:

With probability 50% she gets outcome 0.
With probability 50% she gets outcome 1.

If her outcome was 0, the new wave function is $|00\rangle$
If her outcome was 1, the new wave function is $|11\rangle$

2. Bob now measures in the computational basis

If Alice's outcome was 0, and Bob's outcome will be 0
If Alice's outcome was 1, and Bob's outcome will be 1

Correlation

$$|v\rangle$$



$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Alice's qubit

Bob's qubit



If Alice's outcome was 0, and Bob's outcome will be 0
If Alice's outcome was 1, and Bob's outcome will be 1

Notice that this is NOT signaling. Why?

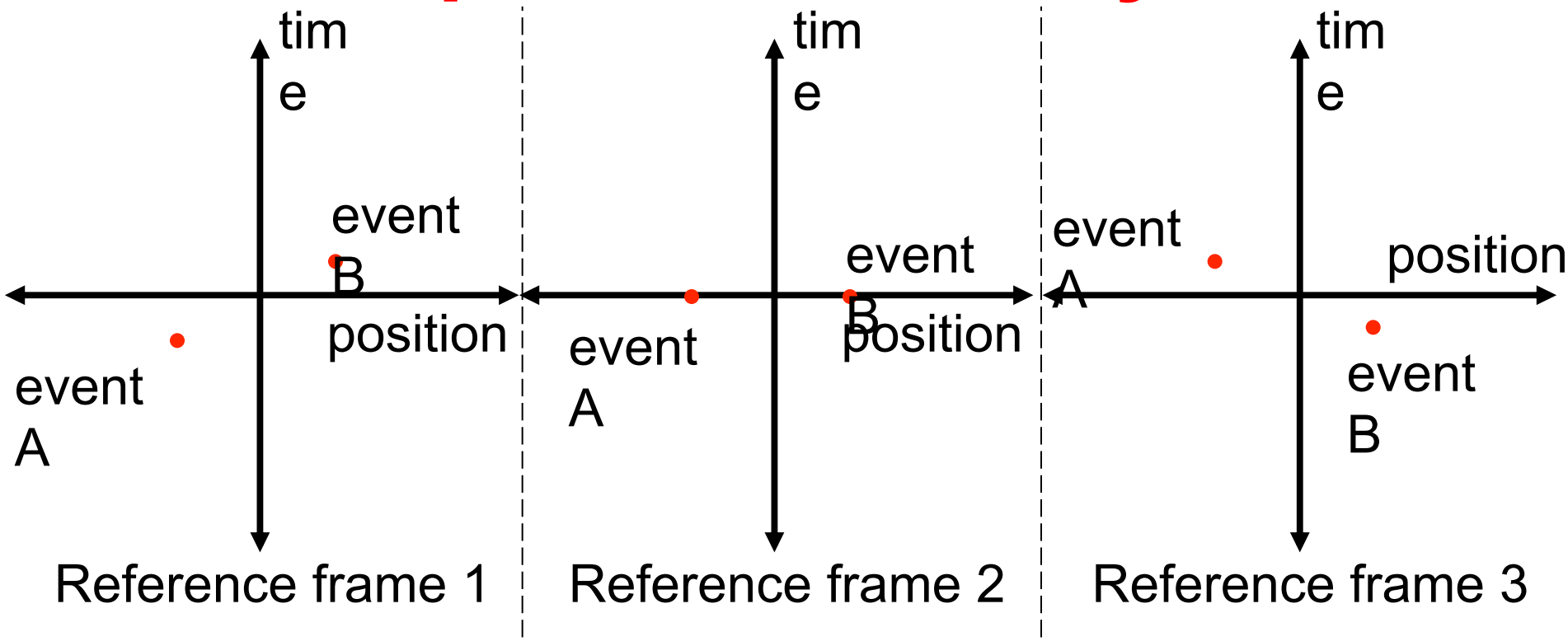
What does Bob see?

Having NOT seen Alice's measurement outcome, he finds that he gets

0 with 50% probability and

1 with 50% probability.

Special Relativity



Should we worry that "who does the measurement first" depends on the frame of reference?

From the perspective of each party, NO, because they always get the same probabilities of outcomes.

Correlation

$$|v\rangle$$





$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$





Alice's qubit

Bob's qubit

A completely classical way to simulate this experiment:

1. Flip a fair coin.
   If the result is heads, put 0 into two boxes.
   If the result is tails, put 1 into two boxes.
2. Give Alice and Bob the boxes.
3. Parties perform measurements by opening their boxes and reporting the classical bit inside as their measurement outcome

Correlation

$$|v\rangle$$





$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$





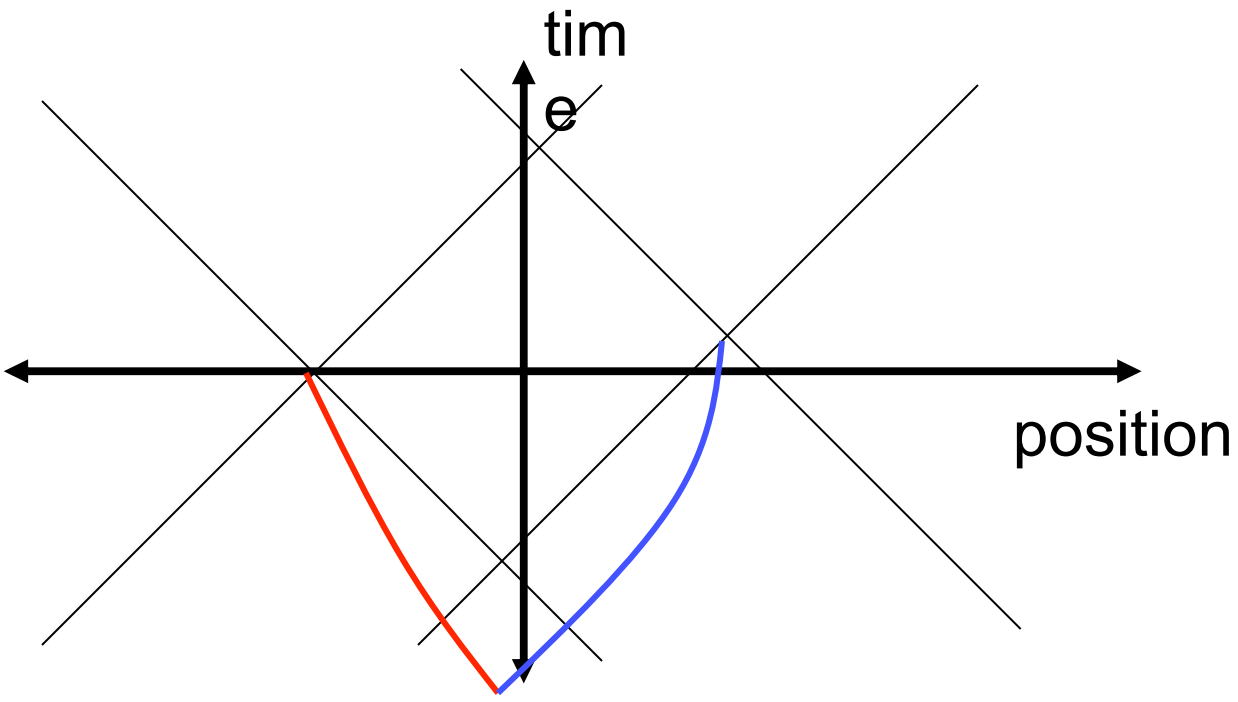
Alice's qubit

Bob's qubit

A completely classical way to simulate this experiment:

1. Flip a fair coin.
   If the result is heads, put 0 into two boxes.
   If the result is tails, put 1 into two boxes.
2. Give Alice and Bob the boxes.
3. Parties perform measurements by opening their boxes and reporting the classical bit inside as their measurement outcome

Local Hidden Variable Model



Suppose we try to respective special relativity, but at the same time, reproduce the probabilities we get from quantum systems

Local Hidden Variable Models

Einstein, Bohr (1927-36)

After the creation of quantum theory, Einstein became very concerned about quantum theory.



Einstein: “God does not play dice.”
Bohr: “Stop telling God what to do.”

Actually Einstein was not troubled by the indeterminism of quantum theory, but by the fact that he thought quantum theory was probably not the “ultimate” theory. He thought quantum theory was incomplete.

John Bell



In 1964, John Bell showed that the question of whether or not quantum theory could be explained by a local hidden variable theory was an EXPERIMENTAL question (and thus a real scientific hypothesis!)

“the most profound discovery of science”

A Game

Alice and Bob are locked away in prison cells and they cannot communicate.





A warden plays the following game.

1. He gives Alice and Bob a slip of paper. Written on each of these papers is the letter S or the letter T.

2. Alice are instructed that at a certain time, they will both be required to shout out either +1 or -1.



A Game



Call Alice's output (+1,-1) if she received S, AS

Call Alice's output (+1,-1) if she received T, AT

Call Bob's output (+1,-1) if she received S, BS

Call Bob's output (+1,-1) if she received T, BT

Alice's slip	Bob'slip	Product of their answers
S	S	ASBS
S	T	ASBT
T	S	ATBS
T	T	ATBT

Note that the product of their answers depends only on what each party does locally.



A Game



The Warden will let Alice and Bob free if they produce

Alice's slip	Bob'slip	Winning Product
S	S	+1
S	T	+1
T	S	+1
T	T	-1

Is it possible for Alice and Bob to always win this game?

No. ASBS= ASBT=+1 implies that BS=BT.

ATBS= -ATBT=+1 implies that BS=-BT.

But this implies BS=BT=0, contradiction!



A Game



Suppose they play this game lots of times, each time they are given one of the sheets of paper with equal probability (i.e. the Warden gives each party an S or a T with 50% probability)

An indication of how well they are doing is to calculate the probability of winning:

Probability of winning= ¼(Pr(ASBS=+1|S,S)+ Pr(ASBT=+1|S,T)
+ Pr(ATBS=+1|T,S)+Pr(ATBT=-1|T,T))

How are they doing?



A Game



Suppose Alice and Bob always play the same strategy.
This means assigning values to AS,AT,BS,and BT.

Probability of winning= ¼(Pr(ASBS=+1|S,S)+ Pr(ASBT=+1|S,T)
+ Pr(ATBS=+1|S,T)+Pr(ATBT=-1|S,T))

At most they can win ¾ of the time.
Why? There must always be a contradiction with the winning formula and so one of these probabilities is zero.

And they can achieve the case where one is zero and all three others are one by always having A and B output +1



A Game

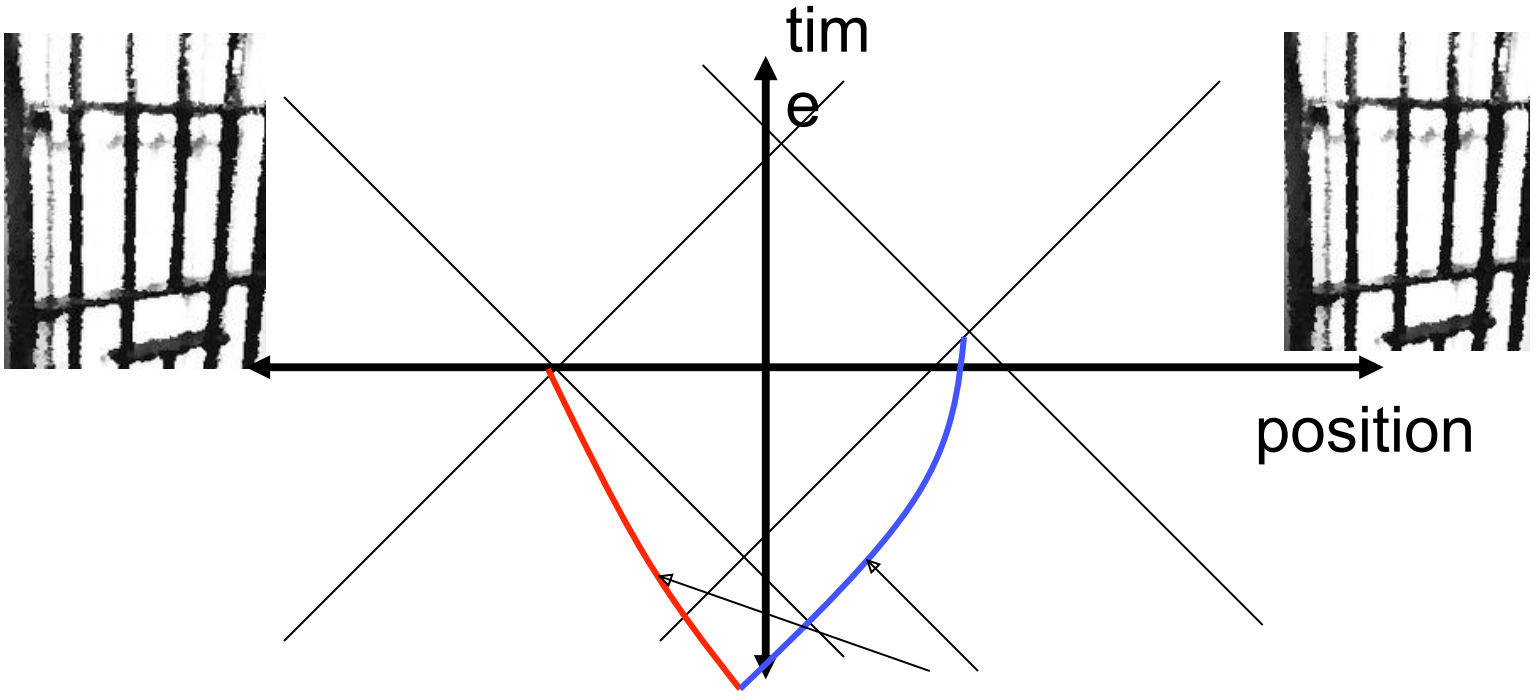


Suppose Alice and Bob are even allowed to share some random numbers they copied from each other before they were thrown in
jail.
What is their maximum probability of winning? Same as before.

Why? Fix the random numbers. Run the protocol.
Probability of winning will be less than ¾.
Average over the random numbers.
No help. Probability of winning is at most ¾.

Probability of winning is at most ¾

The Connection



Local classical data
Local hidden variables



A Game



Suppose Alice and Bob are even allowed to share an entangled quantum state which they made when they were plotting together before the were thrown in jail.

Suppose they each have one qubit of two qubits with wave function

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

A Basis

$$|a(\theta)\rangle = \cos(\theta)|0\rangle + \sin(\theta)|1\rangle$$

$$|b(\theta)\rangle = \sin(\theta)|0\rangle - \cos(\theta)|1\rangle$$

These states are orthogonal

$$\langle b(\theta)|a(\theta)\rangle = \cos(\theta)\sin(\theta) - \sin(\theta)\cos(\theta) = 0$$

And normal

$$\langle a(\theta)|a(\theta)\rangle = \cos(\theta)\cos(\theta) + \sin(\theta)\sin(\theta) = 1$$
$$\langle b(\theta)|b(\theta)\rangle = \sin(\theta)\sin(\theta) + (-\cos(\theta))(-\cos(\theta)) = 1$$

Measurement operators in this basis, labeling these outcomes + and -

$$M_+(\theta) = |a(\theta)\rangle\langle a(\theta)| \qquad M_-(\theta) = |b(\theta)\rangle\langle b(\theta)|$$

A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \langle v|M_+(\theta) \otimes M_+(\phi)|v\rangle$$

$$Pr(+,-) = \langle v|M_+(\theta) \otimes M_-(\phi)|v\rangle$$

$$Pr(-,+) = \langle v|M_-(\theta) \otimes M_+(\phi)|v\rangle$$

$$Pr(-,-) = \langle v|M_-(\theta) \otimes M_-(\phi)|v\rangle$$

A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+,-) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,+) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,-) = \frac{1}{2}\cos^2(\theta - \phi)$$



A Game



They share $|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

If Alice got a slip with S, she measures in the basis with $\theta = \theta_S$
If Alice got a slip with T, she measures in the basis with $\theta = \theta_T$

If Bob got a slip with S, she measures in the basis with $\phi = \phi_S$
If Bob got a slip with T, she measures in the basis with $\phi = \phi_T$

Both parties output their measurement (+ = +1, - =-1)

Choose angles which maximize the probability of winning

A Measurement

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Suppose that Alice measures in the $|a(\theta)\rangle, |b(\theta)\rangle$ basis
and that Bob measures in the $|a(\phi)\rangle, |b(\phi)\rangle$ basis

Probabilities of outcomes:

$$Pr(+,+) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+,-) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,+) = \frac{1}{2}\sin^2(\theta - \phi)$$

$$Pr(-,-) = \frac{1}{2}\cos^2(\theta - \phi)$$

$$Pr(+1) = \cos^2(\theta - \phi)$$

$$Pr(-1) = \sin^2(\theta - \phi)$$



A Game



$$Pr(\text{win}) = \frac{1}{4}\left[\cos^2(\theta_S - \phi_S) + \cos^2(\theta_S - \phi_T) + \cos^2(\theta_T - \phi_S) + \sin^2(\theta_T - \phi_T)\right]$$

$$\theta_S = 0, \theta_T = \frac{\pi}{4} \qquad \phi_S = \frac{\pi}{8}, \phi_T = -\frac{\pi}{8}$$

$$Pr(\text{win}) = \frac{1}{4}\left[\cos^2\left(-\frac{\pi}{8}\right) + \cos^2\left(\frac{\pi}{8}\right) + \cos^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{3\pi}{8}\right)\right]$$

$$Pr(\text{win}) = \frac{1}{2}\left(1 + \frac{1}{\sqrt{2}}\right) \approx 0.85355 > \frac{3}{4}$$



A Game



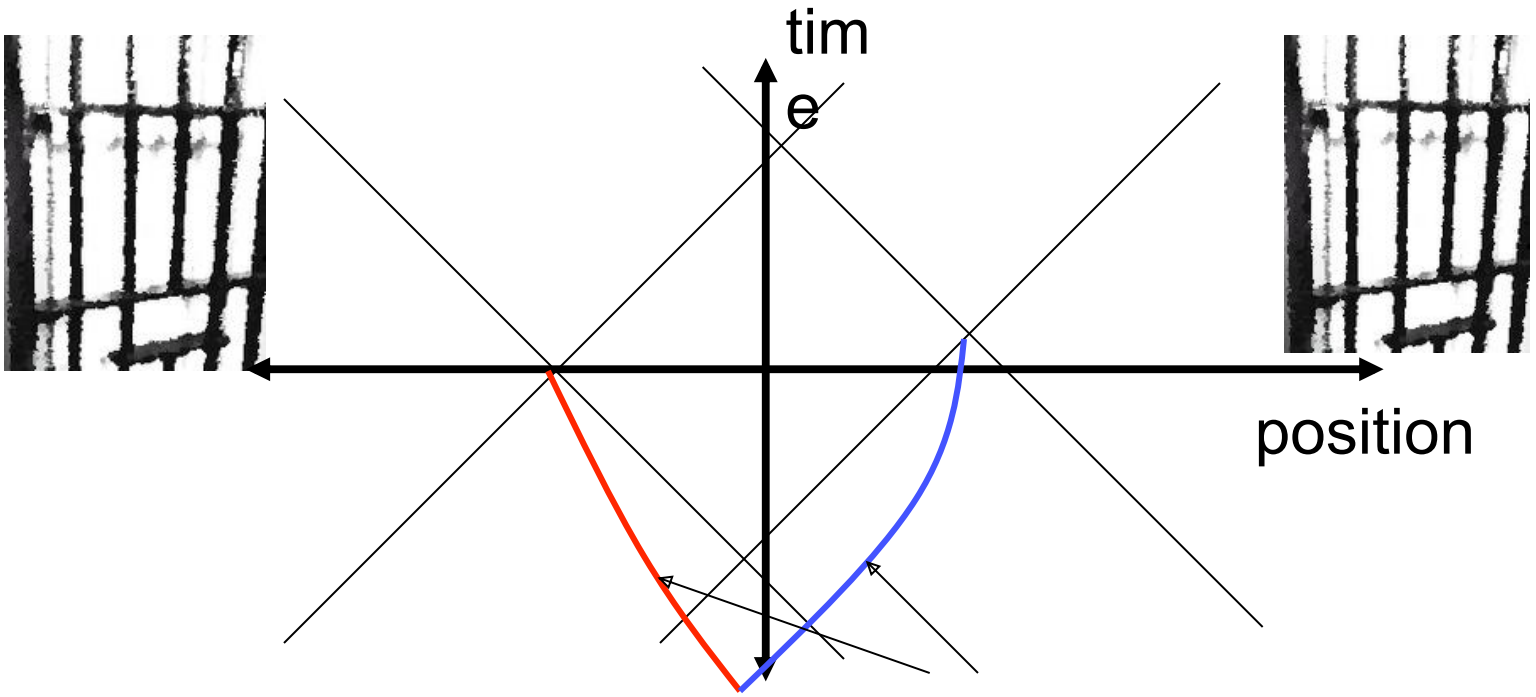
$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

By sharing this entangled state, which they cannot use to communicate with each other, but they can increase their chances of defeating this evil evil warden.

Bell's Theorem

$$|v\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

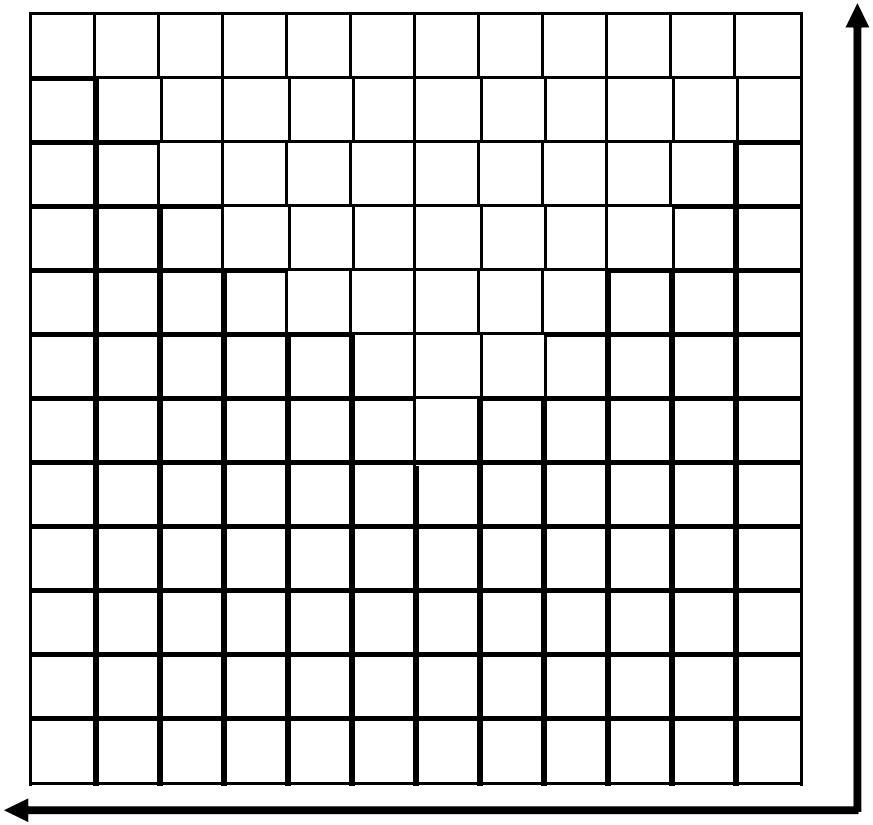
Bell's theorem tells us that we cannot simulate the statistics of quantum theory using a local hidden variable theory!



Local classical data
Local hidden variables

Bell's Theorem

Why is this profound?



Local theories, with classical information, like cellular automata cannot reproduce the predictions of quantum theory

(Apologies to Stephen Wolfram)

Bell's Theorem

Why is this profound?

Bell's theorem is an experiment:

All local hidden variable theories satisfy:

$$Pr(\text{win}) \leq \frac{3}{4}$$

But quantum theory predicts

$$Pr(\text{win}) = \frac{1}{2}(1 + \frac{1}{\sqrt{2}}) > \frac{3}{4}$$

So go out and test it!!!! Famous tests of Aspect 1982.

Experimental demonstration of quantum-correlations over more than 10 kilometers

W. Tittel, J. Brendel, B. Gisin, T. Herzog, H. Zbinden, and N. Gisin
University of Geneva, Group of Applied Physics, 20, Rue de l'Ecole de Médecine, CH-1211 Geneva 4, Switzerland
e-mail: wolfgang.tittel@physics.unige.ch
(June 26, 2005)

Energy and time entangled photons at a wavelength of 1310 nm are produced by parametric down-conversion in a $KNbO_3$ crystal and are sent into all-fiber interferometers using a telecommunications fiber network. The two interferometers of this Franson-type test of the Bell-inequality are located 10.9 km apart from one another. Two-photon fringe visibilities of up to 81.6% are obtained. These strong nonlocal correlations support the nonlocal predictions of quantum mechanics and provide evidence that entanglement between photons can be maintained over long distances.

Bell's Theorem

Why is this profound?

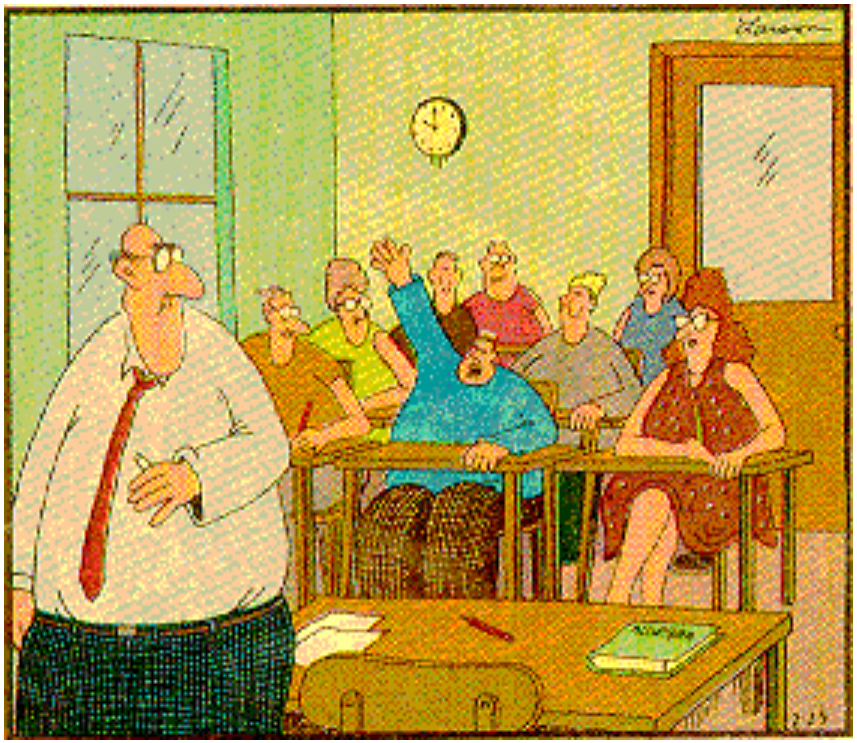
Recall our model where we put bits into boxes to get correlation

Correlation is not profound.

But quantum correlation is different from experiments where we generate correlations locally!

This was truly the first result in quantum information:

quantum correlations play games better
than classical correlations



“Dave, may I be excused? My brain is full.”